

SOCIODEMIC: THE UNSDG OPEN PEDAGOGY FELLOWSHIP PROJECT - THE ASSOCIATIONS BETWEEN GENDER AND RISK OF INFECTIOUS DISEASE



Flora Rudacille & Gracie McDonough
College of Southern Nevada

SOCIODEMICS: The UNSDG Open
Pedagogy Fellowship Project - The
Associations Between Gender and Risk of
Infectious Disease

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UNSDG FELLOWSHIP PROJECT

UNSDG FELLOWSHIP PROJECT FALL 2025

PROF. FLORA RUDACILLE HUMAN BEHAVIOR DEPARTMENT SOCIOLOGY COLLEGE OF SOUTHERN NEVADA LAS VEGAS, NV

BIOGRAPHY

Hi! My name is Professor Flora Rudacille, and I teach Sociology at the College of Southern Nevada – Henderson Campus. I have been teaching at CSN for over 18 years and have taught Sociology significantly longer than that. I attended the University of Pittsburgh for both my B.A. and M.A. in Sociology and have completed the coursework for my PhD. I also earned an M.Ed. from the University of Nevada Las Vegas.

Prior to CSN, I worked for the University of Pittsburgh Office of Child Development. During my tenure at Pitt, I worked in a leadership capacity in several service demonstration and research projects including the Comprehensive Child Development Program, Early Head Start, the MacArthur Foundation CCDP Follow-up Project, and the Pennsylvania Family Centers Initiative.

My professional background and personal experiences make it possible for me to share real-life examples of the application of sociological concepts and methodology with my students. I love nothing more than helping my students to develop their sociological imaginations!

I applied for the UNSDG Fellowship because I have a life-long mission to do as much as I can to promote global and domestic equality and equity for women. Women are far more likely to be unable to read and write, are often prevented from entering the workplace and when they do, earn far less than men. Women are often denied family planning and other women's health services and are much more likely to become victims of physical and sexual violence and are often denied basic medical care.

Our research focus, related to UNSDG Goal #5, Gender Equality, is the connection between epidemiological and sociological factors including gender as they relate to different health conditions and diseases. Medical research has historically focused on men, and even when it has not, the data collected often doesn't allow for the disaggregation needed to reflect the differences in findings for different demographic subgroups. Our project also aligns with other UNSDG Goals, including Good Health (Goal 3), Reduced Inequality (Goal 10) and Clean Water and Sanitation (Goal 6).

My partner, Dr. Catherine Glass of Langara College in Vancouver, Canada, and I, decided to work with our students in connection with UNSDG Goal #5, specifically working with them to examine the association of gender and other socioeconomic factors with disease. We developed three scaffolded assignments, a research proposal, plan and rationale, an annotated bibliography and a research paper. This mini text contains the papers produced by the student groups in my Principles of Sociology (Sociology 101) class.

Professor Flora E. Rudacille College of Southern Nevada - Human Behavior Department Las Vegas, Nevada

UNSDG Biography - Flora Rudacille

UNSDG FELLOWSHIP PROJECT - FALL 2025

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HUMAN BEHAVIOR

DEPARTMENT SOCIOLOGY

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Professor Flora E. Rudacille

College of Southern Nevada - Human Behavior Department

Las Vegas, Nevada

1: PESTICIDES FINAL ACC

Pesticides as a Man-Made Vector of Disease

Daena Tappeiner, Dylan Ulat, Gelly Miravet, Stuart

Henderson College of Southern Nevada

Principles of Sociology Professor

Prof. Flora Rudacille

December 7, 2025

Although not a pathogen in the traditional sense, pesticide exposure functions epidemiologically: it is unequally distributed across populations, structurally patterned in predictable ways, causes both acute and chronic health conditions, and remains widely underreported in global datasets (Boedeker et al., 2020; Donley et al., 2022). Exposure occurs through skin contact, inhalation, or ingestion, producing immediate symptoms as well as long-term latent effects comparable to infectious conditions such as Lyme disease, tuberculosis, hepatitis B/C, and HPV (Gangemi et al., 2016). The burden of pesticide-related illness falls disproportionately on socioeconomically vulnerable populations — agricultural workers, migrants, women, children, and surrounding communities — whose heightened risk reflects broader inequities in labor systems, poverty, regulatory enforcement, and environmental health (Donley et al., 2022; López-Gálvez et al., 2019). Because these disparities map directly onto global commitments to gender equity, labor protections, and public health, pesticide exposure is deeply intertwined with the United Nations Sustainable Development Goals, particularly UNSDG 5: Gender Equality, which emphasizes eliminating gender-based harm and ensuring safe, dignified working conditions for women worldwide. Understanding pesticide exposure as a man-made epidemic — one that mirrors the pathways, transmission patterns, and structural inequalities of traditional infectious diseases — situates this issue within a broader framework of global development, human rights, and cross-national responsibility (UN, 2015). The following sections examine the pathways of exposure, the social and economic stratification that heightens risk for women and migrant agricultural workers, the gendered and biological health effects that follow, and the global policy failures and structural invisibility that allow pesticide-related illness to persist as an overlooked public-health crisis. Taken together, these analyses demonstrate how pesticide exposures function epidemiologically and should be understood as a structural public-health issue that deserves coordinated international action.

Pesticides act as a vector of disease through multiple exposure routes, most notably inhalation, ingestion, and dermal absorption; all of which create both acute and chronic health risks. Agricultural workers are the most directly exposed during mixing, loading, and applying pesticides, but contemporary research shows that exposure extends far beyond active spraying. A large-scale soil analysis across eleven European countries found that 83% of agricultural soils contained pesticide residues, and 62% contained mixtures of two or more chemicals (Silva et al., 2019). Notably, residues were detected even in soils that had not been treated for several years, demonstrating the chemicals' persistence, mobility, and ability to contaminate ecosystems long after their intended use.

These findings show that pesticides function epidemiologically: they persist in environmental “reservoirs,” spread across geographic areas, and create chronic exposure pathways for entire communities, not only for those directly involved in pesticide handling. These residues create chronic exposure risks for entire communities, not just those directly applying the chemicals. Farmers themselves recognize these interconnections. In Indiana, DeBruicker Valliant (2012) documented farmers' awareness that land-use decisions, livestock health, and chemical inputs directly affect local families and the overall community's well-being. Many farmers noted that soil contamination, chemical drift, and livestock exposure directly influence community health, contributing to increased illness and long-term health risks. This perspective reframes pesticide exposure as a public health issue rather than a purely occupational or environmental one, highlighting how its impacts extend beyond individual workers to the broader social and ecological structures that shape agricultural communities.

A major pathway beyond the worksite is the “take-home route,” in which pesticides enter households on clothing, shoes, tools, and skin. A systematic review of more than thirty biomonitoring studies by López-Gálvez et al. (2019) found that children of farmworkers consistently had higher urinary pesticide metabolite levels than non-farmworker children, often two to six times higher, depending on the region and chemical. Environmental sampling studies further show that pesticide residues accumulate in house dust, laundry areas, car seats, and kitchen floors, creating continuous low-level exposure for family members who have never worked in the fields. This pathway demonstrates that pesticide drift extends beyond airborne or environmental routes and becomes socially transmitted, circulating through households and disproportionately affecting children, pregnant family members, and older adults who have never worked in the fields, effectively replicating the transmission patterns of traditional disease.

While economic and socially stratified communities typically face higher risks from pesticide exposures, women are particularly vulnerable. The risks pesticides pose to women's health extend beyond reproductive issues and can also lead to latent illnesses linked to a higher likelihood of chronic disease development. Albadrani et al. (2024) establish a framework connecting pesticide exposure to an elevated risk of breast cancer, reporting that women living in Brazil's dense agricultural zones faced a 60% higher risk compared to those in non-agricultural areas (Panis & Lemos, 2024). Similarly, Shekhar et al. (2024) illustrate a strong relation between pesticide exposure and systemic autoimmune diseases such as rheumatoid arthritis and lupus — conditions that women already face at elevated baseline risk (Parks et al., 2022). These findings also link pesticide exposure to heightened risk for multiple cancers, including breast, ovarian, prostate, leukemia, and lymphoma, underscoring the severe and systemic threat pesticides pose to women's long-term health.

Social position further shapes exposure risk. Migrant women in agricultural work often occupy the most undesirable and hazardous labor roles, frequently because these positions are the only ones available to them within stratified labor markets. Chronic shortages of personal protective equipment (PPE) and even basic amenities such as drinking water ultimately force workers to adopt “bring your own” policies, increasing bodily risk and compounding exposure. Without gloves, masks, or protective clothing, workers' skin, respiratory systems, and eyes receive direct and repeated exposure to harmful chemicals, UV radiation, and dust. This unprotected exposure triggers immediate acute effects, including chemical burns and respiratory irritation, and accumulates over time to produce chronic conditions such as dermatitis, chronic bronchitis, and musculoskeletal degeneration.

Many migrant agricultural workers lack the power to refuse unsafe work or demand protections, and employers often knowingly expose them by disregarding safety protocols. Most cannot afford basic PPE such as gloves, masks, or sunscreen, and endure premature physical deterioration to maintain employment. In this way, the combination of chemical exposure and inadequate workplace conditions becomes a direct source of harm — one rooted not only in environmental hazard but in economic precarity and structural vulnerability.

Globally, pesticide exposure is pervasive, affecting diverse regions including Asia, Europe, and the Americas. In Pakistan, Bakhsh et al. (2016) found that among female cotton pickers, only 1% could correctly identify pesticides or recognize their harmful effects, and only 18–22% were aware of spraying schedules in the fields where they worked. Nearly all women (98–100%) reported transporting contaminated cotton from fields to farmhouses, practices that, coupled with low awareness, heightened exposure to hazardous chemicals (Bakhsh et al., 2016, p. 8). Due to critical gaps in occupational health education about the chemicals these women handle daily, workers are unable to take even simple precautions to protect themselves or prevent contaminated clothing from entering the household.

Similarly, in Spain, Villa-Cordero et al. (2025) documented extreme working conditions for female migrant farmworkers, including shifts of up to ten hours beginning as early as 3 a.m., which left workers with only five to six hours of rest between workdays — despite legal requirements for a 12-hour break. Chronic shortages of personal protective equipment (PPE) and even basic amenities such as drinking water ultimately forced workers to adopt “bring your own” policies, further increasing bodily risk and the potential for harm. The resulting physical strain produced chronic fatigue and impaired cognitive functioning, compounding the risks of an already hazardous environment. Without gloves, masks, or protective clothing, workers' skin, respiratory systems, and eyes were directly exposed to harmful chemicals, UV radiation, and dust. This level of unprotected exposure triggered immediate, acute effects such as chemical burns and respiratory irritation, and accumulated over time to contribute to long-term conditions, including chronic bronchitis, dermatitis, and musculoskeletal damage. Many migrant agricultural workers lack the power to refuse unsafe work or demand protections, and employers often knowingly expose them by disregarding safety protocols. Most cannot afford basic protective equipment such as gloves, masks, or sunscreen, and endure premature physical deterioration to keep their jobs. In this way, the combination of chemical exposure and inadequate workplace conditions becomes a direct source of harm. Additionally, it is worth noting that many of these acute symptoms and latent health effects follow trajectories that closely mirror those of traditionally classified disease vectors, both in form and function.

In Thailand, Lorenz et al. (2012) reported that 11% of pregnant women applied pesticides at work, 51% experienced exposure at home, and 28% personally applied pesticides during pregnancy. Unsafe handling practices risk bringing contaminated clothing into kitchens, bedrooms, or food storage, compounding the risk for both women and their fetuses. During pregnancy, a period of heightened physiological sensitivity, prenatal exposure to organophosphates, carbamates, and organochlorines has been linked to neurotoxicity, as well as cognitive, behavioral, and developmental impairments in the fetus, including birth defects. Unfortunately, the reinforcement for their family's immediate economic survival comes at the expense of their own long-term health, propagating a system of generational harms when coupled with social expectations for women to uphold both domestic responsibilities and

participate in the workforce. Across these regions, exposure intensity is a transnational public health issue, and pesticide-related illnesses constitute a global environmental health crisis.

Despite its widespread health implications, pesticide exposure remains an underrecognized and largely unaddressed global health concern. When examined through the intersecting lenses of gender, labor, and environmental policy, the disparities become particularly stark. As Donley et al. (2022) argue, “People of color and low-income populations are disproportionately exposed to higher levels of pesticides, both occupationally and environmentally, due to systemic racism, economic inequality, and weak regulatory protections” (p. 2). Migrant and agricultural farmworkers, especially within the United States, continue to bear disproportionate harm as a result of regulatory gaps and uneven enforcement. At the global level, Boedeker et al. (2020) estimated approximately 385 million unintentional pesticide poisonings and 11,000 annual deaths. Although this figure has been criticized for relying on extrapolated data, critiques of “overestimation” overlook a central reality: pesticide-related illness is chronically underreported. Given the well-documented weaknesses in surveillance systems — particularly in developing nations where exposure is most severe — Boedeker’s estimate is likely conservative. Studies such as Awaluddin et al. (2021) demonstrate that official reporting systems often capture only a fraction of actual cases, describing underreporting as “a critical barrier to prevention” for workers facing chemical hazards. When these patterns are read alongside earlier evidence of environmental “reservoirs,” take-home exposure, extreme working conditions, and gendered reproductive harms documented in Pakistan, Spain, Thailand, and Brazil, they suggest that what appears in global datasets likely represents only the visible fraction of a much broader, structurally obscured burden of disease.

Policy failures in the U.S. further illustrate how well-documented hazards can persist despite extensive evidence and the existence of agencies specifically tasked with regulating them. As Sellers et al. (2025) describe, “(d)espite multiple risk assessments since the 1990s documenting potential risks to children and farmworkers, chlorpyrifos remained widely used until 2024 when most food and feed uses were finally cancelled” (p. 5). This decades-long delay underscores how political and economic considerations can outweigh public health, even when risks are thoroughly documented. A parallel dynamic appears in Awaluddin et al.’s (2021) analysis of Malaysia’s occupational disease reporting system, where “the underreporting of occupational disease and injury remains a critical barrier to prevention, with agricultural workers particularly vulnerable to pesticide exposure due to inadequate surveillance and poor enforcement of safety regulations” (p. 7). Their findings mirror patterns observed across many rapidly industrializing regions, suggesting that the underreporting documented in Malaysia is not an isolated phenomenon but part of a broader global trend. Taken together, these studies indicate that the unequal distribution of pesticide-related harms observed in the United States likely reflects even deeper disparities in lower-income countries, where weak regulatory infrastructure, limited labor protections, and insufficient enforcement leave high-risk populations (especially women in migrant agricultural roles) effectively invisible within public health systems. In light of the chronic illnesses, reproductive losses, and intergenerational harms documented throughout this paper, this invisibility is not a neutral gap in knowledge; it is a policy decision that sustains working conditions in which those who sustain global food production remain the least protected, the least counted, and the most disposable within existing structures of labor and health governance.

In conclusion, pesticide exposure operates in many ways like an epidemic, producing both acute and latent health effects and spreading through agricultural work, contaminated food and water, and even contact with residue carried on objects and individuals. As with many well-studied infectious diseases, the pathways of exposure run directly through the basic necessities that sustain human life — an especially relevant concern as agriculture remains central to global development and population growth. What becomes clear across the literature is that this is a genuinely global issue, shaping health outcomes in low-income regions and fully industrialized nations alike. While pesticide exposure poses risks to everyone, from curators to consumers, the research indicates that women in migrant agricultural roles are particularly vulnerable, not only because of the occupational hazards embedded in their work, but also due to the culturally constructed social responsibilities that compound their risk and magnify their exposure across both workplace and home. These intersecting biological and socioeconomic vulnerabilities mirror the structural patterns found in traditional epidemiology, underscoring why pesticide-related illness must be understood as a man-made disease vector rather than an isolated environmental hazard. Situating these findings within the framework of the United Nations Sustainable Development Goals (particularly UNSDG 5: Gender Equality) further emphasizes that the global distribution of pesticide-related harm is not merely a matter of environmental exposure, but a reflection of structural inequities that disproportionately endanger women and marginalized labor forces. By treating pesticide exposure through this broader epidemiological lens, we bring necessary visibility to a vitally important public-health issue — one that has remained largely invisible in data and neglected in policy despite its scale. Recognizing pesticide-related illness as both a socioeconomic structural failing and a human rights injustice is not merely an academic exercise; it is a call for meaningful policy

reform, for protections that center marginalized populations, and for a reframing of agricultural progress that does not sacrifice the health of the people who sustain it.

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2: HEART DISEASE FINAL ACC

The Gender Divide in Heart Health
Messiah Bautista, Jewel Lacson, Adam Estrada, Casey Brockett
College of Southern Nevada
SOC 101: Principles of Sociology
UNSDG Goal #5
Professor Flora Rudacille
December 7, 2025

Heart disease is often seen as a “man’s disease.” Still, startling statistics show it has taken nearly as many lives of women as men each year, remaining under-recognized and undertreated among females. Understanding the socioeconomic, biological, and lifestyle factors that contribute to the risk of heart disease amongst women worldwide can benefit future generations. There is an urgent need for gender-specific medical approaches, along with increased awareness, to improve preventative measures that can be implemented globally, aiding in the decrease in women’s risk of heart disease.

Regardless of your location, the biological factors that influence heart complications in the female body should have identifiable implementations that women can benefit from. Women often face different socioeconomic challenges in accessing healthcare or even experiencing higher stress levels from caregiving responsibilities. Other lifestyle factors, such as smoking and diet, also contribute significantly to the risk (Zhang et al., 2021). Women are also susceptible to comorbidities such as obesity and depression (Paluch et al, 2023). Meta-analysis has recognized that, regardless of your location and sociological environment, certain measures should be considered to improve the implementation of corrective action.

Women’s health is influenced by different risk factors, including those related to pregnancy and hormonal changes. Gestational diabetes, which develops during pregnancy, requires careful monitoring and treatment to avoid complications in both the mother and the developing child. It increases the risk of future diabetes in mothers and can cause higher birth weights and delivery complications. Preeclampsia, signaled by high blood pressure and possible organ damage, threatens maternal and fetal health and may require early delivery. Menopause forces cardiovascular changes, increasing the risk of heart disease as a result of the decrease in estrogen levels. Hormonal therapy, commonly used to reduce menopausal symptoms such as hot flashes and mood changes, produces mixed effects on cardiovascular health, making it a strong potential approach to treatment based on specific risk profiles of importance. These essential factors in women’s healthcare call for extensive approaches that highlight wellness throughout the life course.

When looking at the differences in being diagnosed with heart disease for men and women, there are clinical symptoms that affect how long each gender can take to be diagnosed. The difference between the symptoms for men and women is that men tend to show it more easily and earlier. It is easily recognized in men physically. Their symptoms may include severe chest pain, shortness of breath, dizziness, etc. (FALLON, C. K. 2019). As for women, they show symptoms, but they are milder, including light chest pains, nausea, fatigue, and indigestion. This leads to being mistaken for other health issues like anxiety, stress, abdominal issues, and menopause etc. That can result in women being diagnosed later on, getting delayed treatment, or being misdiagnosed for other health issues. The consequences of this can lead to women not receiving the proper urgent care that they need in the earlier stages to prevent or reduce their diagnosed heart disease.

Because of these diagnostic challenges, there are ways heart disease can be prevented and treated so that women do not have to be put at greater risk. There are benefits to women being educated about their bodies, learning about specific symptoms, and applying that knowledge to themselves. This goes along with getting themselves checked regularly at a healthcare clinic or doctor. Healthcare providers should also listen carefully to their concerns, paying attention to even their subtle symptoms and risk factors, as they play a crucial role in figuring out what diagnosis it may be. Women and men tend to get the same treatment for heart disease, like medication, procedures, and lifestyle changes. Although these are beneficial, women have different hormonal effects, and this can affect how their blood vessels may react to treatments. Healthcare providers should be advised to give out personalized treatment as women’s bodies respond differently. Focusing on gender specific treatment, which can improve heart health for both women and men, as it can avoid and may even prevent the worst effects of heart disease.

In the long run, acknowledging heart disease as a significant concern for women conveys that it is crucial to decipher different strategies to consider the many factors that mold health outcomes. Women face more unique challenges than men, from pregnancy related possibilities to de-emphasizing symptoms that predict early detection. And with that, improving education, expanding healthcare access, and encouraging personal care can definitely play an immense part in improving women's heart health. With more substantial awareness and unbiased treatment, the decline of heart disease around the world can lessen, which can result in a stronger and healthier population.

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3: HIV FINAL ACC

HIV Affecting Both Men and Women

Sophia Zagorski, Yeliennis Tamayo, Jamir Akins, Chamy Lam

College of Southern Nevada

Soc 101 - 4001 - 4002 FALL 2025

Professor Flora Rudacille

Human Immunodeficiency Virus (HIV) remains one of the most significant ongoing public health challenges in the world, despite major advances in treatment, prevention, and scientific understanding. More than four decades after its discovery, HIV continues to impact millions of individuals globally, disproportionately affecting certain genders, regions and key populations. As of 2024, an estimated 39-40 million people are living with HIV, and approximately 1.3 million new infections occur annually, with the highest burden found in subSaharan Africa and among marginalized groups in other regions (UNAIDS, 2025; World Health Organization, 2025). Although modern antiretroviral therapy has transformed HIV into a manageable chronic condition and significantly reduced transmission risk, persistent inequities in access to care, socioeconomic barriers, stigma, and biological risk factors continue to drive the epidemic. Understanding the multidimensional nature of HIV, including its biological pathways, social determinants, gender disparities, and evolving treatment landscape is vital for developing effective, equitable public health strategies.

HIV is a virus that attacks the immune system, specifically targeting CD4 T-cells, and weakens the body's ability to fight infections and disease. Without treatment, HIV progresses through stages of acute infection, chronic infection, and eventually AIDS, the most advanced and life-threatening form. The Mayo Clinic (2024) notes that early symptoms may resemble common viral illnesses, with fever, swollen lymph nodes, and sore throat making early detection difficult. Transmission occurs primarily through unprotected sexual contact, sharing of needles, from mother to child during pregnancy or breastfeeding, and less commonly through blood transfusions in settings where screening is inadequate. Although the basic mechanisms of transmission are well understood, the likelihood of transmission is strongly linked to an individual's viral load. Evidence demonstrates that people living with HIV who are on effective antiretroviral therapy (ART) and maintain viral suppression below 200 copies/mL do not sexually transmit the virus, a concept known as U=U (Undetectable = Untransmittable) (Rodger et al., 2019). Systematic reviews further confirm that transmission risk is effectively nonexistent under conditions of sustained viral suppression (Broyles et al., 2023).

While HIV is a global epidemic, its distribution is deeply unequal. Sub-Saharan Africa remains the region most heavily affected, accounting for approximately two-thirds of all global HIV cases (UNAIDS, 2025). Young women and adolescent girls in this region face disproportionately high rates of infection due to a combination of gender inequality, socioeconomic vulnerability, limited access to education, and structural barriers in healthcare. Globally, gendered risk patterns vary based on regional context. In high-income countries such as the United States and Western Europe, men who have sex with men continue to represent the majority of new infections, with 67% of U.S. new diagnoses occurring in this population (HIV.gov, 2025). These differences reflect not only biological risk factors but also cultural norms, stigma, and unequal access to prevention tools. In Eastern Europe and Central Asia, where 94% of new infections occur among people who inject drugs and other key populations, socioeconomic disparities and criminalization contribute to rising incidence (CDC, 2024; UNAIDS, 2025). These regional variations highlight that HIV transmission is shaped by a complex interplay of biological susceptibility, social structures, and public health policy.

Biological differences help explain why women and men experience HIV differently. Women have nearly double the per-act risk of acquiring HIV during vaginal intercourse compared to men due to factors related to genital tract inflammation, sexually transmitted coinfections, and the composition of the vaginal microbiome (Scully, 2018). Hormonal differences further influence immune response, altering susceptibility. In sub-Saharan Africa, high rates of HSV-2 coinfection significantly increase HIV vulnerability because genital inflammation facilitates viral entry. Conversely, men benefit from certain protective factors, such as the reduced risk of HIV acquisition associated with circumcisions, which decreases the amount of susceptible foreskin tissue (Scully, 2018). These biological determinants reveal why gender-specific prevention strategies are essential, especially in regions with limited healthcare access.

Beyond the physical aspects of HIV, the virus also has profound neurological and cognitive implications, even among individuals on effective ART. Research indicates that traditional classifications of HIV-associated neurocognitive disorders (HAND) may overestimate impairment, leading to stigmatization and inaccurate assumptions about cognitive decline. Nightingale et al. (2023) argue for new diagnostic frameworks that account for broader social, medical, and neurobiological factors, emphasizing a shift toward more accurate assessment of HIV-associated brain injury. As HIV-positive individuals live longer due to effective treatment, understanding these long-term outcomes becomes increasingly important for improving quality of life and developing supportive care strategies.

Modern HIV treatment has drastically reshaped the epidemic. Antiretroviral therapy not only restores immune function but also prevents transmission. Recent cohort studies demonstrate that newer ART regimens, such as dolutegravir-based therapy, offer improved viral suppression and retention outcomes but have not always been equitably accessible. Dorward et al. (2023) found that women, particularly young and pregnant women, were less likely to initiate or transition to dolutegravir despite its benefits, revealing structural gender inequities within healthcare systems. Meanwhile, innovative scientific research is exploring the future possibilities of HIV cure strategies. Mathematical modeling studies suggest that sustained viral remission or eradication could substantially reduce HIV incidence, but partial or temporary remission without careful monitoring may increase transmission risk (De Bellis et al., 2025). These findings highlight both the promise and the complexity of cure-oriented research.

Public awareness and education remain essential components of HIV prevention. Despite progress, stigma remains a major barrier worldwide, discouraging individuals from seeking testing, treatment, or preventative measures such as PrEP (pre-exposure prophylaxis). Community-based awareness efforts show that even with declining global HIV rates, millions remain undiagnosed or untreated, and misinformation continues to circulate, particularly among youth and marginalized groups (SEO invigo, 2025; Williams, 2024). The World Health Organization (2025) emphasizes that achieving the global 95-95-95 targets ensuring that 95% of people know their status, 95% of diagnosed individuals receive ART, and 95% of those on ART achieve viral suppression is critical to ending the epidemic by 2030. Expanding prevention tools such as long-acting PrEP, condoms, harm-reduction programs, and widespread testing remains crucial for reducing new infections and addressing disparities.

HIV continues to challenge global health systems, but it is also a disease that can be effectively managed, prevented and controlled with the right strategies. Scientific evidence clearly demonstrates that ART eliminates the risk of sexual transmission when viral suppression is achieved, affirming the power of treatment not only as therapy but also as prevention. However, the persistent gender disparities, regional inequalities, and social determinants of health underscore that biomedical advances alone are not enough. Addressing HIV requires a coordinated global commitment to equitable healthcare access, gender-sensitive prevention strategies, social support, and ongoing public education. As research continues to advance toward the possibility of an HIV cure, the preservative remains to strengthen prevention, reduce stigma, and ensure that all individuals regardless of geography, gender, or socioeconomic status have access to life-saving resources. With a comprehensive and inclusive public health approach, meaningful progress toward ending the HIV epidemic is both possible and within reach.

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4: CHLAMYDIA 1 FINAL ACC

UN-SDG CASE STUDY – CHLAMYDIA TRACHOMATIS and GENDER

Amari Carlson, Payton Powell, Dylan Lujan, Ernesto Robles-Rodriguez

College of Southern Nevada (Henderson Campus)

Professor Rudacille

SOC: 101

12/3/2025

Chlamydia is a type of sexually transmitted infection that mostly affects women. It can cause serious problems if it is not treated, so it is important to understand why it affects women the most. In this project, we are looking at different countries to see how they deal with chlamydia and how the infection rates in women are different around the world. By comparing countries, we can get a better idea of why women are more at risk in some places and how different prevention methods can help.

Chlamydia can be serious for women. “In the USA alone, more than 1.7 million cases are reported each year,” (Qu et al., 2021) and “women also face serious risks like infertility, pelvic inflammatory diseases, and ectopic pregnancy” (Qu et al., 2021) if it is not treated. One reason it is worse for women is that the infection can reach parts of the body that cause long term problems and many women do not notice any symptoms at first. This means that infections can go unnoticed and can cause trouble later. Studies have looked at how often women are checked, how the infection spreads, and what methods work best to prevent it in different countries. Studies show that if chlamydia is not being treated, it can have long-term effects on women’s health. “Early screening and proper treatment can help reduce infection rates and prevent serious health outcomes for women,” (Qu et al., 2021) which shows why focusing on women is so important when trying to control this infection worldwide. Looking at different countries also helps us understand which programs are effective and which ones could be improved to protect women the most.

Research on women’s health in Europe shows that young women experience the highest rates of chlamydia trachomatis infection and face the most serious long-term consequences when cases go untreated. Two major studies highlight both the prevalence of infection and the unique reproductive risks women face. In Germany, Gassowski et al. used data from two population-based surveys and a laboratory surveillance system to measure infection rates across age groups. Their findings show that chlamydia is most common among adolescent and young adult women, with 2.8% of girls aged fifteen to seventeen and 2.3% of women aged eighteen to twenty-four testing positive. The study also noted that men are screened far less often, creating undetected infections that contribute to continued transmission among women. The authors emphasized that early screening is essential to prevent long-term harm.

A Dutch study by Hoenderboom et al. followed women who had previously participated in a national screening trial to examine long-term reproductive outcomes. The researchers found clear links between prior chlamydia infection and pelvic inflammatory disease, ectopic pregnancy, and tubal factor infertility. Because these complications primarily affect women and can result in permanent reproductive damage, the study demonstrates how untreated infections disproportionately impact women’s health.

In comparison, the United States shows similar patterns but takes a different approach to detection and treatment. Chinnock et al. studied the effects of rapid nucleic acid amplification testing (NAAT) in emergency departments, comparing a new 90-minute test to the standard 24-hour turnaround time. After the rapid test was implemented, incorrect treatment rates for women dropped from 37% to 19, while the control group saw almost no improvement. These results indicate that faster testing significantly improves treatment accuracy for women and reduces missed or delayed care. Across both Europe and the United States, young women remain the most affected group, but countries differ in how they address the issue. Europe through population-level monitoring and the U.S. through faster clinical diagnostics.

In Australia, specifically in Melbourne and Sydney, researchers conducted a study that would follow the women up to fifty-six days “post treatment” and determine whether women contracted chlamydia through reinfection or treatment failure. They used various methods, including comprehensive sexual data collection, the detection of Y chromosome DNA, and high discriminatory chlamydial genotyping (Hocking et al, 2013). Moreover, they would measure the azithromycin antibiotic levels of the women using a “validated liquid chromatography - tandem mass spectrometry” method to determine if low azithromycin absorption could be a cause of treatment failure (TF). Finally, the behavior surrounding chlamydia and the lowest dose of azithromycin given to prevent this infection will be experimented with as well.

After this study, they concluded that “if our hypothesis is correct and 8, rather than 23 fail chlamydia treatment with azithromycin, then nearly 3,000 women in Australia and 70,000 women in the USA were inadequately treated for chlamydia in 2011” (Hocking et al, 2013). Moreover, this treatment failure could lead to persistent infection with longer time of each infection and higher risk of complications. Additionally, they also found that there is “little monitoring of chlamydia antibiotic susceptibility, and the role of antimicrobial resistance in treatment failure is unknown” (Hocking et al, 2013). If this study finds that one gram of Azithromycin is sufficient in treating ct, then this will contribute to making treatment of chlamydia more efficient.

There is also relevant information in China. In one study done from 2006-2020, it focused on “newly diagnosed and reported chlamydia cases in women aged 15-49” (Zhao, et al, 2025). They included all diagnosed cases starting January 1, 2006 to December 31, 2020. They included data from twenty-one provinces in Guangdong Province. They used Quasi-Poisson regression models to examine the relationship between sociodemographic factors and chlamydia incidence. Results were that “from 2006 to 2020, 523,367 new chlamydia cases were reported among women of reproductive age in Guangdong” (Zhao, et al, 2025). They found that from 2006 to 2020, the percentage of chlamydia increased exponentially, from 47.4 in 2006 to 95% in 2020. Also, in older women, “9,045 cases were reported, with a mean reported rate of 4.9 per 100,000, rising significantly from 0.01 in 2006 to 9.6 in 2020” (Zhao et al, 2025).

To add on, another study conducted found 3008 patients with infection in thirteen hospitals in twelve provinces of China, which were performed between May 2017 and November 2018. In total, 2,908 participants were included. “The prevalence rates of chlamydia and gonococcal infections in women with genital tract infections were 6.33% and 0.01% respectively” (Zhang et al, 2023). Equally important, premarital sex, first sexual intercourse before age 20, and bacterial vaginosis were “high-risk factors” in their analysis. To end, they concluded that since most chlamydia cases were asymptomatic and no vaccine was available, strategies to combat the infection should include “behavioral interventions as well as early screening programs” to properly identify chlamydia trachomatis and treat individuals efficiently.

Compared to the U.S., and Europe, Australia is not as developed in their methods in how to combat chlamydia trachomatis. However, they are not far behind. When we look at China, their rates are increasing far more and by greater numbers than both the U.S. and Europe. Moreover, their strategies to treat this infection are less advanced. Overall, in combating ct globally, methods/strategies like the U.S. and Europe are leading the way in research and solutions.

In Kenya, specifically Kisumu, Nairobi, and the coastal region a study was conducted by multiple researchers looking at young HIV-negative cis-gender men, (MSM) who often have intercourse with men and transgender women, (TSW). All 650 participants of the study were given 650 Kenyan shillings (U.S \$5.6) for scheduled follow-up visits. At baseline 15.8% of MSM and 27.5% of TSW had chlamydia or gonorrhea, transgender women were also discovered to have higher rates of oropharyngeal CT/NG at around 15.2% compared to cis-gendered men at around 1%, “despite the higher prevalence of infections among TSW there were no differences in CT/ NG prevalence and incidence between TSW and MSM” (Sanders EJ, et al. BMJ Open 2025) ,with both groups getting reinfected at a similar rates even after adjustment, Participants who already had a baseline CT/NG infection were three times more prone to reinfection even well after treatment. Improved contraceptive use, partner notification and newer strategies such as doxycycline post-exposure prophylaxis are needed to help fight the crisis, tests are expensive but important for proper screening and reliable diagnoses especially when dealing with the asymptomatic nature of these infections

This situation highlights a massive unmet need for STI prevention among vulnerable populations who tend to have a lower socioeconomic status.

With all this in mind, it can be concluded that there is still far more research to be done toward combating the prevalence of chlamydia around the world. In countries like the U.S., Europe, and Australia, there seems to be far more research completed already and there are strategies that have been developed and becoming more efficient. In countries like China and Kenya, there still needs to be far more research in developing methods to solve this issue.

Overall, women suffer from this infection far more than men, and in our research, it would be beneficial to include men like women. Sexual behavior and culture do impact the knowledge and ultimately the treatment that women receive, also the time and place they receive it. Women have higher risk for dangerous complications with their pregnancies and overall health when infections go untreated, and developments are underway in combating this. However, we still see that rates are high in women in specific locations, and this is due to socioeconomic status, geographic location, and the efficiency of the nation the women reside in. Despite these disparities, nations are recognizing this as a healthcare problem that needs to be addressed and they are trying to develop better methods, including factors they did not consider before and emphasizing some that they had not, and in doing this they have found success.

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5: CHLAMYDIA 2 FINAL ACC

Chlamydia: Differences between Genders

Haley Horst, Ellie Leatham, Breyia Hee, Luis Brizuela

Sociology Department,

College of Southern Nevada

SOC 101: Principles of Sociology

F. Rudacille

December 9, 2025

In this essay we are going to discuss Chlamydia as it relates to UNSDG Goal #5 to, “Achieve gender equality and empower all women and girls” (United Nations). Chlamydia is a common sexually transmitted infection or STI that is caused by a bacteria called *Chlamydia trachomatis*. It spreads through vaginal, anal, or oral sex with an infected partner. It can also be passed during childbirth. Most people do not experience symptoms, yet the bacteria can still cause damage. When patients do experience symptoms, they are unusually discharge, burning during urination, or pelvic pain. If left untreated, chlamydia can lead to serious complications in men and women. Regular screening is important for young adults, since early detection prevents long-term health consequences. Chlamydia’s relationship to gender is seen by differences in how the infection presents, geographical nature, biological risk factors and statistics.

Studies have shown that the highest infection rate of chlamydia is women aged 15-24, showing more susceptibility. Due to this, it is highly recommended that women under 25 are screened for this disease. The importance is also heightened by the fact that, “70–80% of infections in women and 40–50% in men are asymptomatic” (Sturd & Rucks, 2010). Meaning that almost every case in women is likely to go unnoticed and untreated leading to more longterm health problems. The long-term health problems are often much more severe and can lead to “pelvic inflammatory disease, infertility, and ectopic pregnancy in women” (Sturd & Rucks, 2010). An ectopic pregnancy can then lead to more life-threatening conditions. Research also indicates that antibody responses in women do not prevent ascending into the upper genital tract. Elevated IgG levels “failed to prevent ascension and increased levels of anti-chlamydia IgG were associated with increased risk for incident infection” (Darville, 2019).

Again, Chlamydia trachomatis is one of the most common sexually transmitted infections worldwide, but its impact differs across regions due to variations in healthcare access, awareness, Chlamydia in Women 3 and sexual-network patterns. Even though the organism behaves the same everywhere, the way it spreads and is detected depends heavily on local conditions.

Helble and Sternbach explain that “T-cell responses, especially those involving CD4+ T cells, are central to controlling chlamydia infection” (Helble & Starnbach 2021) . While these immune processes are consistent across populations, the long-term outcomes differ geographically. Areas with limited screening often have more untreated infections, which increases complications and allows the infection to circulate more widely.

Differences in community awareness also contribute to geographic patterns. Lorimer and Hart found that many people in Scotland demonstrated low knowledge about symptoms, testing, and transmission. These gaps often reflect broader regional differences in sexual-health education and access to information. When awareness is low, more infections remain unnoticed, contributing to continued spread.

Chlamydia’s distribution is also shaped by the structure of sexual networks. McMahon showed that certain partner-type patterns create regional “hotspots” where transmission is more likely. By understanding how partnerships form within specific communities, public health programs can target their interventions more efficiently.

Overall, chlamydia is not evenly distributed across the world. Instead, its geographical nature is influenced by differences in healthcare, education, and local relationship patterns. Addressing these regional factors through better screening access, awareness campaigns, and targeted partner-notification strategies is essential for reducing the overall burden of chlamydia.

The next difference is biological risk factors. A biological risk factor is a condition or characteristic inside your body that increases your likelihood of developing a disease. Things like genetics, physiological factors, age, sex, and past or present infections are all biological risk factors. Socioeconomic risk factors are non-medical factors that negatively impact health outcomes. Things like poverty, low and poor education, and limited access to resources. There are many biological and socioeconomic risk factors when it comes to chlamydia.

Understanding who is most at risk begins with looking at the biological risk factors linked to chlamydia. Being a female is one of the biggest biological risks. The cells that line the cervix are especially susceptible to infection. Adolescence and young adults are at higher risk because their cervical cells are not fully mature, making it easier for chlamydia to attach and spread. Having previous sexually transmitted diseases increases biological vulnerability. Together, these biological factors help explain why certain people are more vulnerable to chlamydia infections.

Socioeconomic risk factors play a significant role in the frequency of chlamydia. Individuals with limited access to healthcare may experience the lack of routine screenings, treatment, and sexual health education. All of these increase the likelihood of undiagnosed or untreated infections. Lower income communities often have reduced access to all of these things. Socioeconomic inequality can have an impact on education levels. Low and poor education can lead to gaps in knowledge about STI prevention and safe sex practices. These barriers collectively lead to higher rates of chlamydia. Knowing these socioeconomic barriers is essential for reducing chlamydia rates and promoting good sexual health outcomes.

Chlamydia remains the most widespread and insidious sexually transmitted infection in the world, with adolescents and young adults being the most affected. As one review stated, Chlamydia can persist because "the pathogen inhibits T cell immunity, making it difficult for the host to clear the infection." (Rakesh, 2018). This contributes to immune evasion, which accounts for the tendency of Chlamydia to remain silent-many infected people have no symptoms, complicating the detection and control of this disease.

The epidemiological burden of the infection is enormous. According to one estimate, "more than 90 million individuals" are infected worldwide every year (Satterwhite et al., 2013). This suggests that the problem is global in its scope and that reported cases probably represent only a small fraction of the actual number, given underdiagnosis and underreporting.

From an immunological standpoint, the body does try to fight Chlamydia. According to one review, "T cells restrain Chlamydia trachomatis infections" through their activities, but these are often thwarted when the pathogen interferes with immune responses. Furthermore, the immune response is multifaceted and not always successful in clearing the infection: repeated or low-dose exposures can lead to "aberrant immune responses and enhanced tissue pathology."

Such high prevalence, associated asymptomatic carriage, and imperfect immune clearance combine to make Chlamydia a persistent public health challenge. The fact that millions are infected each year-over 90 million globally-and that many of those infections may go unnoticed, heightens the stakes for robust prevention, screening, and education efforts.

In sum, Chlamydia trachomatis continues to exact a high toll worldwide. Because the bacterium evades the host immune response, most often presents asymptomatically, and infects millions each year, public health approaches must focus on regular screening, particularly in younger, sexually active populations, notification and treatment of sexual partners, and continued research into immunity-based prevention (including vaccines). It is only with this kind of multipronged effort that control, and eventual reduction of Chlamydia's global burden become possible.

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6: CHAGAS DISEASE FINAL ACC

Chagas Disease: Health Risks Across Gender

Elizabeth Meshell, Michael Bowling, Miros Morales Alvarez, Eddie Del Rio Garza

SOC 101 - 4001

Professor Flora Rudacille

December 9th, 2025

Chagas Disease: Health Risks Across Gender

Chagas disease (CD), caused by a parasite called *Trypanosoma cruzi* (*T. cruzi*), is an illness that is most commonly found in Latin America. However, in recent years, there has been an emergence of Chagas disease in other regions of the world, like North America, and even Europe. Chagas disease is gravely endemic in regions where healthcare is lacking and poverty levels are higher. This results in people who have limited medical treatment having more severe complications, making it more difficult to treat without detrimental complications to those already affected by Chagas (Triana, D. R. et al., 2016). Studies have shown that pregnant women often face negative and poor treatments (Avaria, A., & Plaza, C., 2024) and men often face higher risk of heart problems from the disease (Lassen, O et al., 2020). Not only is Chagas disease spreading in multiple new regions, but along with that, there is a concrete impression of inequality when it comes to healthcare between genders and socioeconomic groups.

There has been a plethora of new cases of the disease spreading northward, into the lower United States, along with the disease being carried across borders by infected immigrants. The latter of those two sees a persistent significance on the global scale as the disease can take root in its host and be passed down to children. While originally the parasitic infection was solely native to Latin America, more specifically being endemic to 21 Latin American countries, it is due to a multitude of factors that regions which were once regarded as nonendemic are now brought into question over if *T. cruzi* inflicts the area. Not limited to only bordered regions, kissing bugs—which are the medium for the infection—have been identified in 32 U.S. states (Beatty, et al., 2025). Meanwhile, as for the globalization of the disease itself, it is estimated that 3.5% of Latin American migrants, living in non-endemic countries, had the disease (Nepomuceno, et al., 2024). In other words, about three to four out of every one hundred Latin American migrants across continents such as Europe, Australia, Asia, etc, are carriers of the disease.

Overall, while the migration of kissing bugs up north can be understood from an amalgamation of environmental causes, the migration of Latin Americans who carry CD is subject to patient monitoring and treatment. In a matter of advocacy, Chagas disease necessitates a global concern as its rates in non-endemic countries have substantially increased. To understand from a more quantitative form, “In 2009, it was estimated that about 68,318 to 123,078 immigrants living in Europe were infected by *T. cruzi*,” (Lidani, et al., 2019). With such a global prevalence, the contemporary international rates of the disease warrants some countries to rethink their non-endemic label. Afterall, taking the proper measures to screen and treat for the disease is preferable compared to neglecting it.

Chagas disease is spreading around the world, however, the effects seen from it differ greatly when it comes to gender. Men with CD have worse effects because of the Y-chromosome, most notably posing a higher risk of developing severe heart damage. Studies by Lassen and their team found unique health issues within the male gender: “300 unrelated men who consecutively attended the Hypertension Department of the Córdoba Hospital in the city of Córdoba, Argentina were randomly selected,” (Lassen, O., et al., 2020). While this case-control study was conducted, it was observed that men who did not have the haplogroup called R1b within their Y chromosome, while being infected by Chagas, were five times more likely to develop an enlarged heart. The men were also two and a half times more likely to have additional heart problems. Yet, there is a dilemma, because on average, only four out of ten men had R1b—the Y-chromosome type that dictates that a majority of men on average will see these worsened effects. While chromosome types in males do play a role in putting men at higher risk with cardiac problems, another issue was the “...the inhibitory role of testosterone in humoral responses, particularly regarding TS-specific plasma antibodies in TS+A-vaccinated males. Testosterone depletion in this group enhanced IgG2a- and IgG1-specific responses, with TS-specific IgG2a levels in castrated TS+A-vaccinated mice reaching levels comparable to those in TS+A-vaccinated females...” (Balbi, C.B., S., et al., 2017). Males with normal testosterone levels will on average have reduced antibodies to respond to CD in response to this vaccine. When testosterone is removed, it is observed in male mice that the level of the IgG2a and IgG1 antibodies increases, showing that testosterone limits these antibodies that could help fight Chagas disease.

So, because testosterone suppresses antibodies that could be crucial in fighting back against Chagas, men's "harder" in the sense that their biology makes Chagas disease not only. Not only does testosterone suppress immune responses, but it also makes males more susceptible to getting contaminated by Chagas and makes males have a harder time combating Chagas as well. Considering all this, both genetic and hormonal factors directly influence how Chagas disease affects males and because of this, exclusively men are more at risk when it comes to cardiac complications. Regular testosterone levels, along with the lack of the R1b Y-chromosome in men, also result in men having worse immune system response against *Trypanosoma cruzi*. Not only does testosterone in males lower the combative effects of the disease, but men are also subjected to worse symptoms.

While men are likely to be more susceptible to chagas disease because of genetic and hormonal factors, women face an equally concerning set of health risks specifically regarding vertical transmission. Vertical transmission is where the *Trypanosoma cruzi* parasite is passed from mother to offspring during the gestational period throughout pregnancy. This passageway of transmission calls attention to the importance of screening pregnant women to prevent the spread of congenital CD in infants, as an early diagnosis and treatment would significantly reduce infant mortality (Hinton, S., 2024; Edwards, M., & Montgomery, S., 2021). Studies conducted by Morven Edwards and Susan Montgomery stress how essential pregnancy-based screening is, despite its lacking implementation in endemic regions.

Along with the biological risks, there are societal and health access barriers that further complicate the process for women. Studies in Chile show that pregnant women with Chagas are often misdiagnosed and neglected in healthcare environments, which can lead to the spread of the disease or delayed treatment (Avaria, A., & Plaza, C., 2024). This is due to the lack of gender-sensitive healthcare and socioeconomic discrepancies, which limits women's access to opportune diagnosis and treatment. These issues continue to affect women who reside in low-income housing, where exposure to kissing bugs is higher, leading to a greater risk of infection. These factors highlight the necessity in learning about the severity of Chagas disease and how it affects men and women. This prompts the question of how these gender-based differences influence both the clinical and developmental outcomes of this illness.

Chagas disease is often described as a generalized public health challenge, although research reveals significant sexual dimorphism in how the disease is acquired, transmitted and how it progresses. Understanding these sex specific differences is crucial for effective diagnosis and treatments. Overall, analyzing general prevalence data across endemic and non endemic regions suggests that men and women face similar infection rates, with some studies showing slightly higher infection rates among women in certain immigrant populations (Irish, A.B., Montgomery, S.P. & Marcus, R, 2022). However, this overall equality masks the fundamental differences in risk factors of men living in vector-endemic areas. Men in these areas face a higher risk of infection due to occupational activities involving outdoor and agricultural jobs which involve greater exposure to the kissing bug (Balbi, C.B., S., et al., 2017).

The most critical divergence in transmission methods relates to women's unique role in perpetuating the disease. Congenital transmission from mother to fetus during the pregnancy remains a major pathway of transmission at approximately 4.7% per pregnancy (Balbi, C.B., S., et al., 2017). This only highlights the need for screening women of childbearing age and preventing pediatric cases. Perhaps, the most clinically significant dimorphism lies in the progression and severity. Research focusing on host immune responses strongly suggests that sex hormones influence the severity of Chagas cardiomyopathy, the potentially fatal heart damage caused by the parasite. This signifies that, because women produce estradiol (estrogen), female hosts tend to exhibit greater resistance to severe acute and chronic infection in comparison to men. This is due to the fact that estrogen is linked to an enhanced anti-body mediated immune response and better at managing the disease. On the contrary, the testosterone in men is attributed to increased susceptibility and immunosuppressive effects on humoral immunity which leads to higher risks of severe cardiac complications (Balbi, C.B., S., et al., 2017).

In summary, Chagas disease is not a monolithic condition and its impact on public health varies significantly by sex. While overall infection rates may be comparable, women are distinctively targeted by the critical congenital transmission pathway, and men face a higher physiological risk of severe cardiac progression due to hormonal influences on the immune system. The treatments in the future should consider the dimorphic aspect of this disease and how it transmits. Public health strategies should consider incorporating the development of treatments modalities that account for the hormone driven differences in immune response and severity, specifically in male patients. While studies are still being done, comprehensive screening for pregnant women to eliminate congenital transmission is what is now being recommended.

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7: TUBERCULOSIS 1 FINAL ACC

Gender and Tuberculosis: Global Gender Differences in Risk, Diagnosis, and Outcome

Scott Collier, Hiedi Haneefazi, Au'tiana Powell, Savannah Wagner

College of Southern Nevada Sociology-101

Professor Flora Rudacille

December 7, 2025

Gender and Tuberculosis: Global Gender Differences in Risk, Diagnosis, and Outcome

Tuberculosis (TB) continues to be one of the most significant infectious diseases worldwide, affecting populations across nearly every region and socioeconomic level. According to the World Health Organization's most recent global report, an estimated 10.6 million people developed TB in 2022, demonstrating that the disease remains a persistent threat despite being both preventable and curable (World Health Organization, 2023). However, TB does not affect all people equally. In many countries, substantial gender disparities shape who gets diagnosed, how quickly they receive treatment, and what their long-term outcomes are. These disparities are influenced by a combination of biological susceptibility, social and cultural norms, and unequal access to healthcare. Partners In Health emphasizes that TB's persistence is tied closely to structural inequities such as poverty, limited access to medical services, and gender-based barriers, which disproportionately harm women and marginalized communities (Partners In Health, n.d.). At the same time, evidence from Horton et al. (2016) shows that men have consistently higher rates of pulmonary, smear-positive TB in low- and middle-income countries, suggesting complex interactions between biological and social factors.

Understanding these disparities is essential for global development efforts because gender influences exposure, diagnosis, and treatment outcomes. Sustainable Development Goal (SDG) 5, which promotes gender equality, is directly tied to improving health under SDG 3 (Good Health and Well-being). By examining both biological mechanisms and social determinants, this paper highlights how gender shapes the global TB burden and why equality is central to disease control.

Tuberculosis remains widespread across all regions, but global surveillance shows clear and persistent gendered patterns. The World Health Organization (2023) reports that men account for the majority of TB cases globally, representing approximately 56 percent of all new infections, while women account for about one-third of global cases. Children comprise the remaining share (World Health Organization, 2023). This pattern has been observed for decades. Men develop and report more cases of pulmonary TB, while women are more likely to have cases that go undiagnosed, unreported, or delayed due to barriers in accessing diagnostic services.

A major contributor to these disparities is the difference between infection risk and disease notification. Rickman et al. (2025), in a large population-based meta-analysis of immunoreactivity surveys, found that boys and girls have similar infection risk throughout childhood, and that infection risk remains nearly equal until adolescence. The disparity emerges after puberty, when adult men show significantly higher rates of confirmed infection, suggesting that hormonal and occupational factors contribute to increased susceptibility or exposure.

Evidence from low- and middle-income countries further supports the global pattern. Horton et al. (2016), in a systematic review across 56 surveys, found that men had significantly higher prevalence of smear-positive pulmonary TB, while women's cases were disproportionately under-notified. Structural and social barriers, including stigma and restricted healthcare access, contribute to this underreporting. The Centers for Disease Control and Prevention (2025) also emphasizes that social determinants such as poverty, limited healthcare access, and crowded living conditions intersect with gender, increasing vulnerability for both men and women but in different ways.

Country-level analyses reveal additional gender dynamics. Marçôa et al. (2018) show that male TB notification rates sharply increase after age 20, with the highest incidence occurring among men aged 40–49. Women, by contrast, reach their highest rate in the 20–29 age range, less than half that of men in mid-adulthood. These findings reinforce that gender differences arise not only from biological susceptibility but from differential exposure shaped by social structures.

Biological factors play a significant role in shaping different outcomes of tuberculosis between men and women. Research consistently shows that men are more likely to develop smear-positive pulmonary TB, a pattern supported by findings from Rickman et al. (2025), who reported higher bacterial loads and more advanced disease at diagnosis in males. Hormonal and immune system differences contribute further to this disparity. According to Dabitao and Bishai (2023), sex hormones such as testosterone and estrogen influence immune responses, with testosterone suppressing certain immune functions while estrogen

enhances them. These biological factors become especially relevant after puberty, when hormonal changes intensify. Marçôa et al. (2018) also note that male biological susceptibility increases during adolescence and adulthood. This demonstrates that, although social and environmental factors are important, biological determinants play a critical role in the observed male-female gap in TB.

Social and cultural factors also shape gender disparities in tuberculosis diagnosis and treatment. For many women, stigma, family disapproval, and even social isolation may follow a TB diagnosis, as described by Carwile et al. (2025). Restrictive gender norms can prevent women from seeking timely care, especially in societies where women need permission before accessing health services, which reinforces structural inequalities and worsens health outcomes (Humayun et al., 2022). Men, on the other hand, may delay treatment due to work obligations or fear of appearing weak, according to Muttamba et al. (2024). These social challenges mean that both men and women may end up receiving care at more advanced stages of illness. This aligns with the Centers for Disease Control and Prevention (2025), which highlights social determinants as critical factors driving TB vulnerability.

When comparing TB across different economic and cultural settings, research shows that gender patterns do not always look the same everywhere. In Pakistan, women may appear to have more TB cases in certain regions; however, data is complicated by under-diagnosis and social barriers that prevent women from seeking medical attention (Khan et al., 2013). Similarly, studies in Uganda show that women may encounter cultural barriers that further reduce diagnosis rates and delay treatment because of gender norms that prevent them from seeking care independently (Muttamba et al., 2024).

Regarding socioeconomic factors, research in South Korea demonstrates that people experiencing poverty, crowded housing, and limited insurance coverage face delays in seeking care, which affects TB outcomes across genders (Choi et al., 2023). Conversely, in high-income countries, men still tend to have 5 higher TB rates despite greater access to healthcare, which may be linked to lifestyle factors such as smoking and alcohol use (Peer et al., 2023). These findings show that regional conditions, economic factors, and cultural contexts shape gender differences in TB, reinforcing that TB is not simply a biological disease but also a social one.

Gender equality plays a critical role in global TB control because gender-based barriers determine who gets treatment, who survives, and who is diagnosed. According to the World Health Organization (2023), TB continues to affect men and women worldwide, yet women often face additional obstacles including limited access to health facilities and delayed diagnosis. These findings align with the Centers for Disease Control and Prevention (2025), which highlights that health inconsistencies in TB are commonly driven by social differences, including gender.

SDG 5 promotes gender equality and supports the goals of SDG 3 by addressing unequal access to health and well-being. Evidence from Muttamba et al. (2024) shows that improving diagnosis strategies and increasing outreach can help reduce diagnostic delays, especially for women who encounter structural and cultural barriers. Partners In Health further emphasizes that community-based care and increased access to health services are essential for reducing TB mortality, particularly among underserved populations (Partners In Health, n.d.).

Gender differences in tuberculosis are shaped by a combination of biological risk and unequal access to healthcare resources. Horton et al. (2016) and Rickman et al. (2025) show how lack of access to basic health services and social barriers contribute to delayed diagnosis, higher transmission, and preventable deaths, especially among women. Aligning TB goals with SDG 5 strengthens the goals of SDG 3 because it prioritizes equitable disease prevention and treatment. As Partners In Health states, reducing TB requires addressing inequality rather than treating the disease alone. If gender-based differences are addressed more effectively, health systems can make real progress in reducing tuberculosis on a global scale.

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8: TUBERCULOSIS 2 FINAL ACC

UN-SDG Project – Tuberculosis in Men

Victoria Golab, Lily Hutt, Phoenix Parkhurst, and Lesly Gutierrez

Department of Sociology,

College of Southern Nevada

SOC 101: Principles of Sociology

Professor Flora Rudacille

December 7, 2025

Tuberculosis in Men UN-SDG Project – Tuberculosis in Men

Tuberculosis (TB) is a bacterial infection caused by *Mycobacterium tuberculosis* (Hertz & Schneider, 2019, p. 225). When one thinks of tuberculosis, one often thinks of the past. Perhaps one imagines a 19th century sanatorium or a cowboy dying of this disease on a lonely mountain. However, in the interest of furthering UN-SDG Goal #3, Good Health and Well-Being, this paper will investigate the current prevalence of tuberculosis and where it is most common in the world today (United Nations, n.d.). In 2023, the global incidence of this disease was 134 cases per 100,000 population, a number that has been rising slightly since 2020 (World Health Organization, 2024, p. 2). Global prevalence is difficult to estimate due to disruptions in diagnosis and treatment during the COVID-19 pandemic and lack of available data from many high-burden countries (World Health Organization, 2024, p. 16).

The nature of TB's association with human populations is complex, involving biological, socioeconomic, and behavioral determinants. TB disproportionately affects individuals in low and middle-income countries, where poverty, undernutrition, and overcrowding create favorable conditions for transmission. Men are more frequently diagnosed with TB than women, with a global male-to-female ratio of 1.7:1, suggesting the influence of gender-specific risk factors such as smoking, occupational exposure, social contacts, HIV coinfection, and biological differences (Mason et al., 2017, p. 221). This paper will explore this association in relation to UN-SDG Goal #5, Gender Equality (United Nations, n.d.). The Asia-Pacific region bears the highest TB burden, accounting for nearly 58% of all cases worldwide (Mason et al., 2017, p. 221). However, the association between tuberculosis and male gender appears to exist across all geographical regions.

Tuberculosis is found worldwide, with a significant impact across all regions. TB is most common in areas where poverty, overcrowding, and limited healthcare access are major issues. The highest rates of TB are found in South Asia, Southeast Asia, and sub-Saharan Africa (Dabirao & Bishai, 2023, pp. 149-150). Countries like India, China, Indonesia, the Philippines, Pakistan, and Nigeria report some of the largest numbers of TB cases. These regions do face challenges such as high population density, limited medical resources, and higher rates of HIV, which all increase TB spread. In contrast, TB rates in North America, Western Europe, and Australia are much lower. These areas have stronger healthcare systems, better access to medications, and more effective public health programs. TB cases in these regions are usually connected to travel, immigration from high-TB countries, or people with weakened immune systems. (Mason et al., 2017, p. 221). Eastern Europe and parts of Central Asia also struggle with TB, especially drug-resistant TB, which is harder to treat. Poor treatment completion, limited medical support, and economic instability contribute to these challenges.

Tuberculosis is a serious infectious disease that mainly affects the lungs, and smoking plays a major role in increasing the risk of developing it. It shows that people who smoke are more likely to breathe in TB bacteria and become infected: "...smoking and occupational exposures may put men at highest risk slightly later in life" (Mason et al., 2017, p. 227). Smokers are two to three times more likely to get TB than people who do not smoke. Another important factor is that smoking slows down the recovery from TB. Smokers with TB are more likely to develop complications and may require a longer treatment period. In severe cases, smoking can even increase the chance of death from TB because the body is not strong enough to fight the infection. Interestingly, even experiencing secondhand smoke can play a role in TB (Dabirao & Bishai, 2023, p. 14). People who live around smokers, especially children, are more likely to get TB because the smoke weakens their lungs even though they are not smoking. This shows that smoking does not just affect the individual who smokes; it can also affect everyone around them.

Another reason for higher tuberculosis rates in men is that men are simply more likely to come into contact with the disease. Tuberculosis is most prevalent in lower- and middle-income countries, where men and women often have different social roles. In such cultures, men are more likely to perform jobs that entail higher tuberculosis risk, such as mining (Hertz & Schneider, 2019, p.

226). They also “spend more time outside the home and have more social contacts” (Hertz & Schneider, 2019, p. 226). This is corroborated by Miller et al. (2021), who studied the interaction between social networks and tuberculosis in Kampala, Uganda and found that index cases (that is, participants recently diagnosed with tuberculosis) had more male contacts than controls (p. 7). This suggests that men’s position in their social networks may cause them to be exposed to tuberculosis cases more often than women and therefore become infected more often. In addition, prisons, especially in lower- and middle-income countries, are often hotbeds for tuberculosis (Mason et al., 2017, p. 227). Since men are more likely to be incarcerated than women, this represents another unique way in which this disease touches male individuals. All in all, it seems that men are more likely to find themselves in locations and situations where tuberculosis is prevalent, thus increasing their risk of contracting the illness and contributing to its perceived gender bias.

In addition to the previously mentioned risk factors, HIV infection can activate latent tuberculosis infection (LTBI), meaning that HIV and tuberculosis comorbidity is common (Hertz & Schneider, 2019, p. 237). LTBI is carried by ¼ of the population worldwide, but only ever develops into a more active type of TB if the immune system is deeply compromised. HIV tends to be one of the most common causes of weakened immune systems that leads to TB, and the common co-infection rate of HIV and TB has led to a stigmatization of TB in less developed countries.

Though tuberculosis is more prevalent among men, it is important, in the interest of UNSDG Goal #5, Gender Equality, to consider the unique impacts this disease has on women (United Nations, n.d.). Despite its higher prevalence in men, tuberculosis is “among the top five causes of female death in the world” (Hertz & Schneider, 2019, p. 226). Certain groups of women are particularly susceptible to tuberculosis. For instance, in a meta-analysis of seven high-income countries, tuberculosis rates were higher among boys and men in most age groups; however, at ages ten to fourteen, rates were significantly higher among girls (Peer et al., 2023, p. 10). The reasons for this are unclear, although the authors suggest that fluctuation in sex hormone levels during puberty may play a role (Peer et al., 2023, p. 12). Women in their reproductive years are also particularly susceptible to the illness because “pregnancy increases the risk of Tb disease progression” (Hertz & Schneider, 2019, p. 226). Additionally, women in southern and central sub-Saharan Africa experience a greater burden of tuberculosis than men due to unsafe sex and intimate partner violence, which increase their risk for HIV, and, due to the comorbidity of the two diseases, tuberculosis (Rowley & Mugala, 2022, p. 155). Finally, it is interesting to note that women are more vulnerable than men to extrapulmonary tuberculosis (EPTB), or tuberculosis affecting tissues outside of the lungs, though the reasons for this are unknown; for instance, one study in Ethiopia said the M:F ratio was as high as 1:2.9 (Dabitao & Bishai, 2023, pp. 149-150).

Women being affected by EPTB more than men is only one of many abnormalities pointing to the fact that TB has gender specific biological factors that we are not yet aware of. While sociocultural factors do certainly play a role, they alone cannot explain the gender Tuberculosis in Men 6 difference in TB cases. Even in animals, males are more susceptible (Dabitao & Bishai, 2023, p. 160). With EPTB being so common in women and pulmonary tuberculosis (PTB) being mostly in men, things just do not add up with the current scientific understanding. Current knowledge suggests that hormones may play a role, but it is too early to say anything for certain (Dabitao & Bishai, 2023, p. 162).

Overall tuberculosis remains a major global health challenge shaped by a complex combination of biological, social, and gender-based factors. Although the disease affects men at higher rates worldwide, this pattern cannot be explained by social and behavioral determinants alone. Biological influences possibly including hormonal differences likely contribute as well, although these mechanisms remain poorly understood (Dabitao & Bishai, 2023). At the same time, women face unique risks, especially those related to puberty, pregnancy, exposure to HIV, and higher risk of extrapulmonary TB (Hertz & Schneider, 2019; Peer et al., 2023; Rowley & Mugala, 2022). The continued interaction between tuberculosis and HIV further complicates global control efforts, as HIV infection increases the likelihood that latent tuberculosis will progress to active disease (Hertz & Schneider, 2019). Taken together, these factors show that reducing the global burden of tuberculosis requires addressing both the social conditions that promote transmission and the biological factors that shape different risks for men and women. Strengthening prevention, improving access to diagnosis and treatment, and pursuing research into sex-based biological differences are all essential to advancing UN SDG Goal #3, Good Health and Well Being, and Goal #5, Gender Equality (United Nations, n.d.).

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9: SCHISTOSOMIASIS FINAL ACC

Gender Inequality and the Risk of Schistosomiasis

Krystal Ward, Ingri Garcia-Guerrero & Anonymous

Human Behavior Department

Sociology 101 – 4001

Professor Flora Rudacille

December 7, 2025

Schistosomiasis is not only a neglected tropical disease but also a mirror of global inequities. The scholarship in this text demonstrates how biology, environment, and social structures converge to shape its impact. From neurological complications and pregnancy risks to untreated preschool children and gender vulnerabilities, the research highlights how schistosomiasis disproportionately affects marginalized populations. Climate change, migration, and poverty expand its reach, while limited healthcare access and entrenched inequalities deepen its consequences. Together, these sources provide a foundation for examining schistosomiasis as both a medical condition and a sociological issue, showing how human behavior, systemic barriers, and global health policies intersect in shaping outcomes.

Schistosomiasis is more than a parasitic infection; it is a story of how gender roles, social expectations, and biology intersect to shape health. Both women and men are affected, but the ways they encounter the disease are not the same. In Pakwach District, Uganda, Ssali et al. (2023) reveal how women's daily routines like collecting water, washing clothes, and caring for families, place them in direct contact with contaminated sources. These tasks, shaped by cultural expectations, make exposure almost unavoidable. Yet when it comes to treatment, women often face barriers such as stigma, reduced autonomy, and limited access to healthcare. Female genital schistosomiasis (FGS) adds another dimension: Sturt et al. (2025) show that FGS lesions increase susceptibility to human papillomavirus (HPV), which can progress to cervical precancer. Despite these risks, FGS remains underdiagnosed, reflecting broader patterns of invisibility in women's health.

Men's risks often stem from occupational exposure. Fishing, farming, and other waterbased labor places them in frequent contact with contaminated environments. Ssali et al. (2023) note that men are more likely to access mass drug administration programs, but this advantage is shaped by cultural norms that prioritize men's participation in public health initiatives. At the same time, expectations of toughness or self-reliance can delay men's care-seeking, worsening outcomes. Vadukoot et al. (2025) highlights hepatic schistosomiasis in non-endemic regions, where men's migration for work often increases exposure, though both genders face diagnostic challenges when awareness is low.

Schistosomiasis affects both genders, but in gender-specific ways. Women's risks are tied to domestic responsibilities and reproductive health, while men's risks are linked to occupational labor and cultural expectations. Women face consequences such as FGS and its connection to HPV and cervical cancer, while men often experience chronic hepatic complications. Men may access treatment more readily, but women encounter systemic barriers. In non-endemic regions, both genders face diagnostic delays, yet women migrants often experience additional inequities. These comparisons show that schistosomiasis is not simply a medical condition; it reflects how gender shapes health outcomes.

Schistosomiasis is a disease written in the rhythms of daily life, and those rhythms are gendered. Women's exposure through domestic labor and reproductive health risks, and men's exposure through occupational labor and cultural norms reveals how the same parasite creates different vulnerabilities. Addressing schistosomiasis requires interventions that see both sides: empowering women to access care, supporting men to overcome cultural barriers, and ensuring that health systems recognize the inequities that shape outcomes. By situating schistosomiasis within gendered frameworks, we gain not only a fuller understanding of the disease but also a clearer picture of the social structures that shape health across communities.

The spread of schistosomiasis is heavily influenced by its geography. Specifically, areas with warm and fresh water are where these parasites thrive, along with places that do not have easy access to healthcare or safe water. Areas with heavily occurring cases of schistosomiasis consist of sub-Saharan Africa, the Middle East, Latin America, and Southeast Asia. These areas house an ideal environment for schistosomiasis to occur (Colley et al., 2014). Additionally, everchanging factors such as climate change can alter global distribution at the hands of temperature changes, rainfall patterns, etc. (Asare et al., 2025). Globally, schistosomiasis is a significantly neglected tropical disease. The transmission of this disease is driven by limited access to healthcare, lack of water sanitation, and lack of control program coverage (Colley et al., 2010). As certain environments interact with constant

socioeconomic inequalities, the global burden of schistosomiasis is likely to stay within regions of low public health effort, less environment management, and less control strategies.

In differing environmental and sociological contexts, schistosomiasis can lead to intestinal, urogenital, or even neurological complications (Alanazi et al., 2017). For example, pregnant women and children suffer hardest from this disease. Pre-school children from Lake Victoria, for instance, show higher infection rates due to repetitive contact from domestic responsibilities (Nalugwa et al., 2015). Within pregnant women, urogenital schistosomiasis sometimes leads to factors such as low-birth weight in infants, highlighting possible reproductive and developmental implications (Mombo-Ngoma, et al., 2017). From the Lake Victoria example and studies on pregnant women that suffer from schistosomiasis, we see that the disease not only poses risks from biological sustainability, but also from societal inequities that can lead to varying complications.

Africa's climate is ideal for the reproduction and spread of snails; it also creates conditions that make schistosomiasis a persistent public-health challenge. A significant challenge Africa faces is poverty, exacerbated by poor economic policies. This disease is most common in areas with inadequate living conditions that promote poor access to clean water, sanitation, and hygiene. The lack of sufficient domestic funding for control programs and inadequate mapping of disease and snail distribution often leads to a decline in the effectiveness of control strategies over time. Healthcare in many African countries faces significant challenges, including limited access to diagnostic and treatment methods, a shortage of healthcare professionals, and underperforming health facilities (Omohwovo et al., 2024). Another major problem is that most people lack sufficient knowledge and understanding of risk factors for transmission and preventive measures.

An estimated 40 million women of childbearing age suffer from schistosomiasis. Pregnant women attending antenatal clinics in Fougamou and Lambaréné, Gabon, were consecutively screened for the presence of schistosoma haematobium eggs in diurnal urine samples (Mombo- 7 Ngoma et al., 2017). Maternal and newborn characteristics assessed at delivery were compared between infected and uninfected mothers. Urogenital schistosomiasis was diagnosed in 103 (9%) of 1115 pregnant women. Low birth weight was more common amongst infants of *S. haematobium*-infected mothers (Honkpehedji et al., 2017). This association was unaffected by controlling demographic characteristics, gestational age, and *Plasmodium falciparum* infection status. Other risk factors associated with low-birth-weight delivery were underweight mothers.

Schistosoma haematobium causes significant morbidity and may even lead to lifethreatening complications due to its predilection for the female urogenital tract. *Schistosoma haematobium* eggs cause granulomatous inflammation and potential obstruction in the urinary bladder, ureter, uterus, fallopian tube, and ovaries (Qunhua et al., 2000). Studies evaluating the association of *S. mansoni* with pregnancy outcomes have demonstrated an increased risk for anemia, preterm deliveries, and low birth weight infants (Siegrist, Siegrist-Obimpeh, 1992, Ajanga et al., 2006).

The community of people in Lake Albert has been greatly affected by this disease in many ways. People gather their resources from the water, but they cannot use contaminated water. Lake water is used for various domestic purposes, including washing, drinking, and cooking. There have been changes in lake resources and their utilization over time, which are affecting 8 local economies and ways of life. Almost all focus group discussion (FGD) participants across categories from both sites agreed that fish stocks have greatly reduced over time and that it is a critical issue. Overfishing, poor fishing methods, illegal fishing gear, population growth, and climate changes (such as the lack of rain) were given as reasons for the decrease in fish stocks (Anyolitho et al., 2025).

People are also affected by high prices due to inflation. Rising prices have reduced the profitability of snail harvesting and fishing, causing external buyers to stop purchasing these commodities. Participants suggested that minimizing contact with the lake could help address the problem of schistosomiasis in their communities (Neema et al., 2025). There was consensus across communities regarding the need for alternative water sources. The communities proposed that the government should provide alternatives such as boreholes and tap water. This would reduce their reliance on the lake for domestic water needs, thus minimizing contact and potentially solving the problem of schistosomiasis in their communities.

Overall, schistosomiasis is not just a parasitic disease. It reflects socioeconomic intersections of gender, poverty, and inequality. Women and children suffer the hardest due to social roles and systemic neglect, not biological susceptibility. From an outwards perspective, we find that eliminating this disease takes more than just medical practice. It requires resolving inequities within societies. The study of schistosomiasis goes beyond exercise and creates a call to action for global health and gender equality.

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10: EBOLA FINAL ACC

United Nations Project: Beyond the Virus: Examining the Gender and Social Dimensions of Ebola

Lilly Young, Ariana Lemus Cevantes, and Destiny Maland

College of Southern Nevada (Henderson Campus)

4002 - Sociology 101

Dr. Flora Rudacille

December 07, 2025

Examining the Gender and Social Dimensions of Ebola

Ebola disease remains one of the most serious viral threats in global health because of its rapid spread, high fatality rate, and its ability to severely disrupt communities. While the disease is known for its medical severity, recent outbreaks have shown that Ebola is shaped by many social and environmental factors that influence who becomes exposed, who receives care, and who recovers. During the 2022 Sudan ebolavirus outbreak in Uganda, transmission spread across several districts and affected both healthcare workers and community members, reflecting weaknesses in early detection and infection control (Kiggundu, 2022). Gender roles and social responsibilities also play a major part in exposure, as caregiving, household duties, and burial practices can place women and men at different levels of risk (Kabami et al., 2024). Long after the initial illness ends, many survivors continue to struggle with physical symptoms, stigma, and practical challenges that affect their daily lives and ability to work (Tozay et al., 2020). To fully understand the impact of Ebola, it is important to view it not only as a medical condition but also as a biosocial issue influenced by gender and community roles. The rest of this paper will explore these patterns, beginning with an overview of the disease and its clinical features.

Ebola disease is caused by several viruses within the *Orthoebolavirus* genus, including the Ebola virus, Sudan virus, and Bundibugyo virus (World Health Organization, 2025). Early symptoms usually begin suddenly and include fever, intense fatigue, muscle aches, headache, and sore throat. As the illness progresses, patients often develop vomiting, diarrhea, abdominal pain, and signs of kidney or liver involvement. Although Ebola is often thought of as a hemorrhagic disease, bleeding does not occur in all cases and usually appears later in the illness (World Health Organization, 2025). These early symptoms can look similar to malaria or typhoid, which can delay diagnosis and contribute to continued spread. This was seen during the 2022 outbreak in Uganda, where some initial cases had non-specific symptoms like fever and stomach pain that made early identification difficult (Kiggundu, 2022). Understanding these signs and symptoms provides essential context for examining how Ebola impacts different groups and how communities respond.

The Ebola epidemic revealed obvious gender disparities in its impact. Women have been shown to be more susceptible to the virus than men are. Over half of those who become ill are women, and over half of the mortality rates from the virus are also women (Onyeneko et al., 2023). Women's increased exposure can be attributed to the fact that they take on more caretaking responsibilities in the house, while men usually have more outdoor responsibilities and are home less often. Although there is no evidence of biological differences between females or males that increase the vulnerability to the Ebola virus, there are definitely different levels of exposure between the two. It is also important to take into consideration that vulnerability to the Ebola virus is more based upon gender and gender roles, and not a person's sex. For example, women are more inclined to nurse children and care for their sick family members, and it becomes a responsibility as well as part of their gender role (Nkangu et al., 2017). In the 1976 outbreak, the mortality rate was 56% among women and 44% among men. Similarly, with the 1995 outbreak, 53% were in women, while 44% were in men. Even in the most recent outbreak in 2014, there were many more cases recorded among women than men. In many outbreaks, the transmission rate is higher within households than in hospitals, meaning the main caretaker for the household is the most at risk for the virus (Nkangu et al., 2017). Therefore, it is proven that women are at a higher risk of exposure due to their gender roles.

Studies show that women report more self protective behaviors and lower Ebola virus disease vaccine acceptance, while men reported more formal care-seeking as well as a higher Ebola virus disease vaccine acceptance. Men have also reported higher Ebola knowledge and a lower belief in rumors about Ebola. This helps to explain why women continue to get the virus more often and have higher mortality rates than men (Pham et al., 2022). During the seventh Ebola outbreak in the DRC, 74% of the deceased were women. This is an alarmingly high percentage given that there were only 69 patients, 36 being women, and 33 being men. This shows that even if there are similar amounts of men and women being affected, women still have higher mortality rates. Not only

does the Ebola virus disease cause a higher mortality rate in women, but it can also affect women's basic health. It has caused many women to lose confidence in the healthcare system. During outbreaks of the virus, reproductive healthcare services dropped by 30% in some countries. Maternal health and child reproduction also decreased during Ebola outbreaks (Onyeneho et al., 2023). Effects in maternal and reproductive care in areas where outbreaks are common can cause health issues for generations to come, which is something that should be focused on more often when looking at the virus (Schmidt-Sane et al., 2021). The Ebola virus disease affects women in many different ways that people may not expect, and epidemic responses should address the intersection between the Ebola virus and gender to see improvements and lessen cases in women.

Biosocial aspects of the Ebola virus are often disregarded; these aspects of the disease affect people's lives in contracting and surviving Ebola. These aspects can be economic, political, communal, and familial. Taking a look at the economic and political factors of this disease, Sierra Leone is a great example. The country was originally a reparation country for former slaves in 1787 and gained independence from Britain in 1961. 1991 to 2002 Sierra Leone Civil War, and the Kono District was one of the major theaters during the war (Richardson et al., 2016). The Civil War caused 70,000 deaths and displaced 2.6 million, many of whom were across the borders of Guinea and Liberia (Richardson et al., 2016). When Ebola started to break out, many of the displaced people in the Kono district contracted the disease, and then were transported to the Kenema hospital. One of the biggest problems with the disease in Sierra Leone is the fact that there is a lack of medical care and services; the government institutions are generally impoverished (Richardson et al., 2016). The politics in this country is ruled under patrimonial powers, where politicians control through systems of personal relationships that bring policies and favors in exchange for political support. This leads to counterproductive economic development as money often goes to the elite (Richardson et al., 2016). The irresponsibility in Sierra Leone's funds is reflected in their healthcare system, as patrimonial powers lead to the underfunding of community health, which results in serious disease outbreaks.

Communal and familial aspects affect individuals with both contracting the disease and surviving the disease. The familial aspect often falls into how many individuals contract Ebola. In Sierra Leone, many people would take care of the sick in their household; they'd wash sick relatives' soiled clothes and bedsheets, prepare, serve, and feed them food and drinks, and wipe sweat off their bodies (Jalloh, 2024). Although the nation publicized the dangers of taking care of ill relatives. Many disregarded the message due to the common sociocultural practice of taking care of ill relatives. Many people also feared that if the health authorities were to take their sick family member to a treatment center that they would die there and wouldn't get the deceased family member's body back (Jalloh, 2024). The sociocultural values and fear of the health authorities enhance the transmission of Ebola among family members.

There is great fear in contracting Ebola, but even after surviving, the fear of Ebola still lingers. In Liberia, there is a great stigmatization of Ebola, which has resulted in many survivors not disclosing their status. People often suffer from social isolation from their families and communities, physical violence, and job loss. The Liberian Ministry of Health described those who survived Ebola as those who "tested positive for the Ebola virus disease and after receiving care and treatment, recovered and tested negative" (Mayrhuber et al., 2017). Although Liberia has this definition of an Ebola survivor, there has been poor record keeping in many facilities, which makes it difficult to obtain a certificate of survival. Survivors have also stated having long-term physical and psychological effects. Some physical effects include: blindness, hearing impairments, pain in joints, muscles, or back, abdominal pain, changes in the immune system, uncontrollable erections in men, and failure to conceive in women. (Mayrhuber et al., 2017). Many individuals who have survived Ebola have been evicted, lost their jobs, and prohibited from using the community wells. Even some marketers refuse to allow Ebola survivors to buy food because they refuse to accept their money (Mayrhuber et al., 2017). Survivors of Ebola do not just face the sickness of the disease, but the long term physical effects and the stigmatization from their own community.

The impact of Ebola should not only be seen as a horrid medical condition, but should also be acknowledged for its biosocial issue affecting women and communities. Ebola is a serious viral disease that rapidly spreads through close contact with individuals who have contracted the disease. Gender and social roles play a major part in the contraction of Ebola, mainly affecting women. Women have more contact with the disease due to their caretaking responsibilities. Even after surviving the disease, the social stigma of Ebola affects survivors' daily lives physically, psychologically, and socially. The effects of Ebola do not just harm individuals while sick, but even after illness, they still live their day-to-day lives, taunted by the disease.

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11: PULMONARY HYPERTENSION FINAL ACC

Pulmonary Hypertension

Kaleilani Dossett, Sara Mandelke, Zachary Moss & Anonymous

College of Southern Nevada

Sociology 101

Flora Rudacille

December 7th, 2025

Definition

Pulmonary Hypertension Definition Pulmonary Hypertension (PH) is a type of high blood pressure that affects the arteries in the lungs and right side of the heart. Pulmonary Hypertension is a silent but serious disease that often goes unnoticed until it becomes life-threatening. It causes you to have high blood pressure in your pulmonary arteries, which carry blood from your heart to your lungs. This disease usually progresses over time, as the pressure damages your arteries, causing them to narrow and restrict blood flow. Once this happens, your lower right heart chamber must work harder to pump blood through the arteries to reach your lungs. The extra work causes your right ventricle to get bigger from the overwork it's doing, which can lead to right-sided heart failure. If you don't treat Pulmonary Hypertension, it can be fatal.

Pulmonary Hypertension is categorized into five different groups based on its cause. Group 1 is the tissue damage in the pulmonary arteries. It is commonly called Pulmonary Arterial Hypertension (PAH). Group 2 occurs when the left side of the heart is not functioning properly. Group 3 is when you start to get problems with your lung functioning. Group 4 is caused by blockages in the pulmonary arteries, often from blood clots, which is known as Chronic Thromboembolic Pulmonary Hypertension (CTEPH). Lastly, Group 5 covers health problems that are connected to PH in ways experts still do not completely understand. Groups 2 and 3 are the most common types. Group 1 (PAH) is rare, but many people mix it up with PH in general. This is why some think Pulmonary Hypertension is rare, but it actually is not. Experts say PH affects about 1 in 100 people worldwide. It is most often seen in adults over 65, but it can happen at any age, even in newborns and young adults (Cleveland Clinic, 2025) .

Discovery

The discovery of Pulmonary Hypertension is often overlooked, as many of the early signs and symptoms are similar to those caused by other circulatory or respiratory diseases, such as congenital heart disease, asthma, COPD, and sleep apnea. Vague symptoms are often associated with early Pulmonary Hypertension. The initial symptoms of many patients are shortness of breath, exertion, lack of exercise tolerance, fatigue, or chest pain (Ley et al., 2023, pp. 823-824). Pulmonary Hypertension is difficult to diagnose with a routine physical checkup and often goes unnoticed until the underlying symptoms worsen. If a person suspects that they may have a form of Pulmonary Hypertension, the best way to assess the suspicion is via an echocardiogram, which is an ultrasound exam of the heart that provides images to assess any possible abnormalities of the heart and also estimate the pulmonary artery pressure and resistance (Cordina, 2019). There are other tests that can be conducted, such as blood tests, chest x-rays, and electrocardiograms, to check for complications and other lung conditions that are known to cause this disease. Despite the wide array of tests available to check for disorders that may cause Pulmonary Hypertension, there is only one test that can allow for the diagnosis. The gold standard test to accurately diagnose it is with a right heart catheterization. RHC allows for the proper categorization and diagnosis of Pulmonary Hypertension based on the 5 main groups.

Symptoms

PH symptoms usually start mild, making them easy to overlook, but they tend to get worse as the condition progresses. No matter which type of PH someone has, many of the early warning signs are similar. People may notice chest pain or discomfort, dizziness or even fainting, and a constant feeling of fatigue that makes daily tasks harder. Loss of appetite is also common, along with shortness of breath that may first appear during activity and later even at rest. As PH continues to develop, swelling in the feet, ankles, legs, or belly can occur due to the strain placed on the heart and lungs. These symptoms together can greatly affect a person's quality of life and signal the need for medical care.

Pulmonary Hypertension (PH) is categorized into four main stages, called "functional classes," which describe the severity of symptoms and how they affect daily life. In Class 1, people have no noticeable symptoms and can do all normal activities without

difficulty. Class 2 means that symptoms, such as shortness of breath or fatigue, appear during normal everyday activities like cleaning, shopping, or going up the stairs, but the person feels fine while resting. In Class 3, even simple daily tasks become very challenging, causing fatigue or shortness of breath to occur more easily, although rest still provides relief. Class 4 is the most severe stage, where symptoms are present even at rest and worsen with any physical activity, making it extremely difficult to perform normal everyday tasks. As PH progresses through these classes, symptoms become more obvious and disruptive, highlighting the importance of early diagnosis and careful management.

Genetic Passing

Pulmonary Hypertension can be passed down through family genetics, depending on the type of hypertension a person may have. One example of this includes Familial Pulmonary Arterial Hypertension (FPAH). FPAH is harder to diagnose and find due to the disease often skipping generations. FPAH is often caused due to a mutation in the bone protein called bone morphogenetic protein receptor 2 or BMPR2, which affects the artery walls in the lungs (Pha, 2023).

In most cases of FPAH, the patient may have one single wrong copy from a parent's gene, otherwise known as autosomal dominant inheritance. This one faulty gene can increase one's risk of the disease, but is not an automatic indicator or diagnosis that someone may have Familial Pulmonary Arterial Hypertension. When no harm is done, this is known as incomplete penetrance, where the mutation never develops into a symptomatic issue. In very rare cases, a child may inherit two faulty genes from their parents, known as autosomal recessive inheritance, causing the chances of developing the disease to skyrocket.

BMPR2, or the bone morphogenetic protein type 2 receptor, plays a crucial role in the cell walls and how they work to repair themselves. According to the Pulmonary Hypertension Association (n.d.) (PHA), changes in BMPR2 are the cause of nearly 70% of Familial Pulmonary Arterial Hypertension cases. Gender plays a small role in development as well. While around 40% of women with a BMPR2 mutation are diagnosed with PAH, only 10-20% of men are diagnosed (Pha, 2023). While BMPR2 is the most common gene associated with the disease, there are many other genes, such as KCNK3, SMAD9, ALK1, CAV1, and ENG, that can all cause different types of pulmonary hypertension in a person (Pha, 2023). The Pulmonary Hypertension Association (n.d) mentions that there are around 14 known gene mutations that are connected to pulmonary hypertension.

Due to the risk of passing this trait onto your offspring, many doctors advise genetic testing to be done on those who have been diagnosed with some of the gene mutations related to the disease. Genetic tests may include a medical professional receiving a sample of saliva, blood, or even a cheek swab. Furthermore, getting genetic testing done can show a patient important information, including things such as identifying any disease-causing variants present, explaining how a person may have developed PH, and assess if other family members may be at risk for developing the disease.

The Silent Killer

Pulmonary Hypertension is often referred to as “The Silent Killer” due to its slow and subtle developing nature, as well as its non-specific symptoms that can often be mistaken for other issues. Most patients often report experiencing symptoms including asthma, tiredness, and even mild chest discomfort. Due to the vagueness of these symptoms, in most cases it takes physicians months to accurately diagnose the patient's disease. This delay in diagnosis further puts back the patient's care, resulting in worsening symptoms. According to a REVEAL Registry published by the National Library of Medicine, more than 20% of pulmonary arterial hypertension patients go over two years of symptomatic problems before they are accurately diagnosed and treated (Brown, 2011).

Another common reason the disease is nicknamed that way has to do with its difficulty in accurately confirming the disease is found, even when clear symptoms are shown. Pulmonary Hypertension is best diagnosed by performing a right-heart catheterization, an extremely invasive procedure that many patients do not wish to undergo. Although there are non-invasive procedures to go through, including echocardiography, they are less effective at properly diagnosing PH and often miss early cases of the disease. Because of this, patients are not properly treated until the disease is far more advanced. If left untreated for too long, PH can cause irreversible damage to a patient and could ultimately lead to death (Pulmonary Arterial Hypertension).

Non-invasive Procedures

Non-invasive procedures to help deal with Pulmonary Hypertension are often highly favored over surgical procedures. Some typical treatments often include treatments that are meant to be inhaled by the patient, otherwise known as oxygen treatments. This treatment significantly helps to open the patients' airways. Patients living with Pulmonary Hypertension are put on a regular flow of oxygen to help support the lungs with breathing. The oxygen is used all hours of the day and night. The use of blood thinners can also help with the blood flow, including ones such as Warfarin, Jantoven. In addition, Vasodilators, Water pills, and High-dose

Calcium blockers are all other common treatments medical physicians may put their patients on. Fortunately, non-invasive treatments are there to help, along with other treatments and surgeries. All the treatments and medications work hand in hand to improve the quality of life for those who suffer from this disease. Another thing to remember is that some medications and treatments are not yet approved, and those left with Pulmonary Hypertension are left with the “what ifs” that have no answers. Let’s take a look at all the different treatments and medications that will improve the lives of those living with Pulmonary Hypertension.

Surgical Procedures

The most important invasive test is the right heart catheterization, which is used to confirm Pulmonary Hypertension and distinguish the WHO categories. In this process, a catheter is inserted into a central vein through the pulmonary artery and the right heart. It is applied to measure pressures and cardiac output (Mandras et al., 2020, pp. 1982-1983). The results of hemodynamics in this study contribute to the choice of a patient to administer vasodilator medications, surgery, or other invasive treatments (Ryan et al., 2012, pp. 112-114).

Medications

Pulmonary Hypertension is misdiagnosed more than you could imagine. On average, it takes a patient three different doctors before getting a proper diagnosis. Medications to relax blood vessels are called Vasodilators. These medications help open blood vessels and improve blood flow. It is given continuously through a small pump attached to the body. Some of these medications consist of: Epoprostenol (Flolan, Veletri), Treprostinil (Remodulin, Tyvaso), Iloprost, and Selexipay (Upravi). To go a little more in depth, there is the Remodulin pump- this device delivers the medication (Treprostinil) via continuous subcutaneous or intravenous infusion. This is one of the treatments my niece was on in May of 2016, the procedure was performed by Surgeon Dr Nathan Taggart, MD, at St Mary’s Mayo Clinic Rochester MN. Soluble guanylate cyclase stimulators are for relaxing the pulmonary arteries, which helps lower the pressure in the lungs. Riociguat (Adempas) is not to be taken while pregnant. Medications such as Bosentan (Tracleer, Macitentan (Opsumit) and Ambrisentan (Letairis) are all used to widen blood vessels. They are Endothelin receptor antagonists. To increase the blood flow, you are going to take some Sildenafil (Revatio, Viagra) and Tadalafil (Adcirca, Alyq, Cialis)- these medications are also used to treat erectile dysfunction. Let’s move on to Calcium blockers like Amlodipine (Norvasc), Diltiazem (Cardizem, Tiazac). Although effective, a small number of patients with pulmonary hypertension improved while taking them. Nitric oxide dilates blood vessels in the lungs, reducing pulmonary arterial pressure and improving oxygenation. In 2019, my niece was in this study. Inhaled treatments consist of Treprostinil dry inhalation powder (Tyvaso DPI), Inhaled Treprostinil (Tyvaso), and Yutrepia (Treprostinil) inhalation powder (Garaygordobil, 2025).

Treatments

Getting down to the root cause of Pulmonary Hypertension instead of just treating the symptoms is what researchers at the Mayo Clinic and other facilities are working on. Patients who are living with PH are more than likely taking medications that are just managing their symptoms, but soon they may have other options at their fingertips, as researchers are working on developing medications that will target the actual root cause. According to Kelsey Stalvey, Pharm. D., and Zoe Owrutsky, Ph.D, this paradigm shift will offer hope in better outcomes and a longer survival rate, also medically reviewed by Diego Araiza Garaygordobil, M.D. One point in the study is for the use of Sotatercept (FDA approved in 2024), which is a game-changer for those who suffer from PH. The more others are aware of this disease, the more information can be collected, and the more treatments can be available, which in turn can save more lives. Of course, while we sometimes take for granted the simple breaths of air we take, some people living with Pulmonary Hypertension use pure oxygen therapy to be able to breathe. Also, think about when you have a cold or the flu, if you have Pulmonary Hypertension and you are on certain medications, you can’t just grab any off the shelf, otherwise known as OTC (over the counter) medicines, many may have negative side effects to the medications you are already on. Did you know that the FDA has only approved two medications- Bosentan, a receptor, and Sildenafil, an inhibitor, for kids living with Pulmonary Hypertension? Let’s talk about Subcutaneous and Intravenous treatments- for the delivery of medications like Treprostinil, which is just under the skin using a tiny tube and portable pump with 24/7 delivery than that is Subcutaneous. Now, delivering medication directly to a large vein in the chest via a catheter that was surgically placed would be Intravenous treatment. This example is another treatment my niece had endured, called the Hickman line. This is an intravenous treatment surgically inserted into the neck or chest. It is also used for giving and taking blood-infused medications and other treatments such as Chemotherapy or Stem Cell Transplant. Surgical procedures that are possible if medications do not help, to name a few: Atrial Septostomy, Lung or Heart Transplant, Potts Shunt, and the Reversed Potts Shunt. The Reversed Potts Shunt is a procedure my niece had in 2017, performed by Surgeon Dr Sameh Mahmoud Mo Said, M.D. Potts shunt creates a connection between the pulmonary artery to the descending

aorta (pop-off valve)- the tissue is taken from another part of the body and used to gap the connection between the pulmonary artery to the lower aorta, allowing the oxygen-poor blood to bypass the lungs and directly into the systemic circulation. What about during the procedure: - First, the patient is given a sedative to help relax them; It is common for patients to be awake during a Right Heart Catheterization. Catheter- thin, flexible tubes that are placed by a doctor in the neck or groin area, which are then inserted into a blood vessel. The doctor then guides the catheters to the heart. Recovery times vary. Stem cells are not approved yet and are still being researched. While still unproven and experimental, it has been showing potential in animal models; there has been a reduction in inflammation, which promotes vascular repair, and it inhibits the disease's progression (Huang et al., 2016). Therapeutic benefits of induced pluripotent stem cells in Monocrotaline-induced Pulmonary Arterial Hypertension. While Pulmonary Hypertension only affects 1 in 100 people globally, it is crucial to remember the countless numbers of doctors, therapies, and money that goes into living with the disease. Although there is not a lot of awareness about this disease, clinical studies are on the rise to give a better understanding to readers. Dr Zannos Grekos, an assistant clinical professor of Cardiology at Nova Southeastern University and a member of the international team that developed the stem cell treatment protocol. “It goes against traditional theory that we should try to fix the existing pulmonary vasculature, but we are generating new blood vessels with impressive results!” Despite all the available treatments out there today, Pulmonary Hypertension remains incurable, with the main focus being on managing symptoms and improving quality of life. Pediatric Pulmonary Hypertension has been understudied, and little is known, which creates a limited understanding of the disease.

Types of PH

Pulmonary Hypertension is a persistent increase in blood pressure within the arteries of the lungs. The World Health Organization divides it into five types according to cause and the pattern of hemodynamics (Mandras, 2020). (WHO group 1)(PAH) Pulmonary Arterial Hypertension is a disease characterized by the constrictions and obstructive remodeling of the pulmonary vascular bed. Pulmonary Arterial Hypertension is divided into seven subgroups determined by cause: Idiopathic PAH (group 1.1), Heritable PAH (group 1.2), drug and toxin induced PAH (group 1.3), PAH associated with various conditions including connective tissue diseases, HIV infection, portal hypertension, and congenital heart disease (group 1.4), PAH in long term responders to calcium channel blockers (group 1.5), PAH with venous/ capillary involvement (group 1.6), and persistent PH of the newborn (group 1.7) (Mandras, 2020). (WHO group 2)(PH-LHD) Pulmonary Hypertension due to left-sided heart disease occurs in response to an increase in left atrial pressure and is usually a consequence of an underlying cardiac disorder. (WHO group 3)(PH-CLD) Pulmonary Hypertension due to chronic lung disease can occur in many lung diseases, including COPD, interstitial lung disease, and sleep disordered breathing. (WHO group 4)(CTEPH) Chronic Thromboembolic Pulmonary Hypertension is characterized by obstruction of the pulmonary vasculature by organized thromboembolic material and vascular remodeling, resulting from prior pulmonary embolism. CTEPH is likely undiagnosed, and its incidence and prevalence have not recently been established. (WHO group 5) Group 5 is for patients with unclear and or multifactorial causes. The group is composed of four subgroups based on primary cause. (group 5.1) consists of chronic hemolytic anemia or myeloproliferative disorders. (group 5.2) is primarily caused by systemic and metabolic disorders such as glycogen storage disease, sarcoidosis, pulmonary Langerhans cell histiocytosis, lymphangioleiomyomatosis, and neurofibromatosis. (group 5.3) Pulmonary Hypertension associated with fibrosing mediastinitis or chronic renal failure forms. (group 5.4) is associated with a complex congenital heart disease (Mandras, 2020).

Aftermath

Although there are no currently known cures for Pulmonary Hypertension, it is common for patients to be able to live a somewhat comfortable life with the disease. Aspects of life such as physical mobility, energy, sleep, and emotional well-being are all affected significantly due to PH. Because PH affects the ability to exert oneself, even routine tasks can become difficult: walking, climbing stairs, household chores, or shopping may take much more effort or become impractical. A European-wide survey of patients and caregivers found that over half described “very significant” impacts on daily life — including limitations in physical activity, self-care, and domestic responsibilities.

Although it is possible to live with the harsh disease of Pulmonary Hypertension, it is not easy. Many patients have to live with lifelong problems due to their own bodies failing them. The loneliness and discomfort the disease brings to people create many mental problems for patients as well. Patients with PH often feel symptoms of guilt, depression, and fear of the future. With the funding for more research and time spent on finding a cure for Pulmonary Hypertension, there is a chance to help save up to 70 million people from having to deal with this life-threatening disease.

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12: UTI 1 FINAL ACC

Urinary Tract Infection Among Women

Cassidie Higbee, Toni Nigos, and Anonymous

Department of Sociology,

College of Southern Nevada

SOC 101: Introduction to Sociology

Professor Flora E. Rudacille

December 7, 2025

Even though UTI's can be transmitted into men. We can regularly see that this disease is more prominent in women. The reason being is simply because of the biology of a woman's body over a man's body. There are certain biological factors in a woman that make them more vulnerable to getting a UTI in their system of urinary tracts. For example, according to studies from (Springer Nature, 2025) a woman's urethra is shorter, which overall makes the urinary tract system in their bodies more vulnerable to any kind of bacteria to enter that part of their system. One of the main ways it can be transmitted in that way is usually through sexual-intercourse, also certain hygiene products in that area that are used excessively. Because of the fact that women's biological factors are weaker to bacteria, these causes and many more can transmit diseases causing UTIs more often for men. The age group that can be most affected by this disease would be elderly women. Because not only are their urinary tracts already weak, but because of their age. They are experiencing lower estrogen levels due to menopause. Which can cause the breaking down of the urinary tract as a whole, making the cause for UTIs even more vulnerable for them. However it can actually be harder to identify a UTI in older women. Because of the fact that they have risks of higher bacteria in their urinary tract. It overcomplicates identifying what symptoms are from a UTI, and what symptoms are from other bacterias that are in the urinary tract. According to (MDPI, 2025) studies normally UTIs are defined by something called a dipstick. The dipstick is supposed to soak up your urine, and it detects bacteria called leukocyte and nitrites in the urine. But it is specifically meant for the younger generation, and the results that are for older patients tend to not be as accurate. There is also a secondary infection called the recurrent urinary tract infection (RUTI). Which is basically where you get a UTI consistently, specifically 1 every 12-13 months. According to (Andrew Flower, 2014), when this disease is consistent it can give us a higher risk of health problems which leads to an increase in the cost of healthcare for the individual. It can start to impact our daily lives in ways that we wouldn't expect. For example it can cause irritability, tiredness, and finding it harder to concentrate. Which can disrupt a bunch of important things in the day life of a woman.

Infectious disease research expands across both bacterial and viral areas, and the articles referred illustrate the challenges faced in this specific clinical practice. It examines the recent developments in understanding UTIS highlighting the growing complications presented by antibiotic resistance and prostate related risk-factors. This shifts the focus toward HIV prevention, mentioning how young gay and bisexual men evaluate different exposure preexposure prophylaxis (PrEP) delivery methods and how their preferences shape real world strategies that may be prevented. Although bringing these issues together, analyzing UTI management specifically in men in which a population bacterial and viral concerns interrelate.

Urinary tract infections (UTIs) in men despite being less common than in women as they present clear challenges due to urogenital anatomy and frequent involvement of the prostate. According to a review covering advances in UTI and bacterial prostatitis, male risk factors include prostate enlargement BPH (benign prostatic hyperplasia) and urological procedures such as transrectal prostate biopsy. The article also emphasizes the antiseptic properties of prostate secretions, including a potential defensive role for PSA (prostate-specific antigen) which may be key to preventing recurrent infections, highlighting a biological mechanism beyond just basic structural risk factors.

The management of UTIs and prostatitis in men is increasingly complicated by rising antibiotic resistance. In this report it notes underlining that the growth of fluoroquinolone resistance that's linked to more common post-biopsy infections and recurrent UTIs, according to the same analysis suggesting that traditional antibiotic based treatments may not be adequate going forward in the future. In the article it quotes "The increasing global antibiotic resistance also significantly affects management UTI in men and calls for alternative strategies."(Wagenlehner P.2, 2014) The author recommends that such studies investigate different strategies such as focusing on bacterial reservoirs in the mucosa of the bladder wall or other new therapies guided by a deeper understanding of the disease. However some believe a few participants expressed worries about potential negative effects and reduced safety if you miss prescriptions, pointing out the need for clear communication and secure information.

The last study “Urinary tract infections in male veterans with human immunodeficiency virus” examined if the outcomes of treatment or patterns for UTIs differ between male veterans with or without HIV. It’s analyzed “we utilized 2009 Veterans Affairs (VA) administrative data to compare the treatment duration and outcome for men.” (Caitlin P.3,) This importantly described that shorter therapy defined as 7 days or less did not lead to increased recurrence compared with longer courses, it suggests that longer treatment may not always be necessary. These findings challenge any assumption that men with HIV require antibiotics for regimens in UTIs.

The field of infectious disease faces a more complicated environment that is molded by both enduring bacteria risks and developing virus avoidance methods. It demonstrates how prostate-related vulnerability and growing antibiotic resistance continue to make treating infections in men more difficult. It’s shown that medical treatments are only effective when they’re consistent with the patient’s needs and lifestyle, illustrating the importance of adapting preventive measures to real-world behaviors. All these resources highlight the need for patient-centered and scientifically informed treatment of the diseases and the advanced knowledge of human behavior. Healthcare systems may only address the problems that present viral diseases through using a comprehensive policy.

The text highlights The use of biomarkers, describing it as “a characteristic that is objectively measured and evaluated as an indicator of normal biological processes, pathogenic processes, or pharmacological responses to a therapeutic intervention” Using these devices in screenings, diagnosis, or monitoring the disease’s activity as a whole. Additionally they may help Clinicians select appropriate Therapies or gauge therapeutic responses. UTI’s rank among one of the most prevalent bacterial infections among females, therefore the use of Biomarkers is crucial in helping define the issue and provide treatment. Despite an extraordinary amount of research, the identification of new UTI biomarkers has remained elusive, and very little recent research, if any, has translated into something valuable for patient care. This is partly because of the unique challenges that complicate UTI biomarker development. The text emphasizes the need to address the utilization of these devices to help improve the UTI diagnosis process.

This article describes how recurrent urinary tract infections are a problem affecting both women and men, and how there has been experiments conducted that prove different strains might help prevent them. For example there is an experiment conducted in which the effects of pravastatin was assessed on adults with ongoing UTI prescriptions. An analysis was conducted Using trials among participants randomized to receive pravastatin, fosinopril or a placebo over a course of 4 years. The data shows that, “pravastatin was associated with a reduced total number of UTI antibiotic prescriptions (relative risk, 0.43; 95% CI, 0.21–0.88) and occurrence of second UTI antibiotic prescriptions [hazard ratio (HR), 0.25; 95% CI, 0.08–0.77]” as well as, “. Fosinopril was associated with an increased occurrence of first UTI antibiotic prescriptions (HR, 1.82; 95% CI, 1.16–2.88).” This shows that People taking pravastatin had 57% fewer antibiotic prescriptions as well as among the people who had already had one UTI in a biotic prescription those taking pravastatin or 75% less likely to need a second prescription over time. the people taking Faustino prayer over 82% more likely to need an antibiotic for a UTI compared with people not taking it. Considering the data the study suggests Strains exert a higher effect on recurrent UTIs than on first UTIs.

Within this text the author goes into detail about how UTI’s are transmitted and the effects they can have on the body, especially on post-menopausal women. The text highlights the fact that menopausal women are a significant Demographic of those who have been affected by UTI’S, representing a considerable proportion of the adult female population within the U.S. Going into detail about this understudied group, the study aims to quantify the risk of UTIs within them specifically. For example Group Health organizations have involved themselves in the study in which they have used cox proportional Hazard models to analyze the risk factors in order to reach a better outcome for these women. Results within one of the first studies state, “ testing positive for any of the five sexually transmitted infections was 12 times more common among female enrollees ages 15-24 compared to women £55 (1020.41 versus 85.44 per 100,000 respectively).” This concludes that sexual intercourse is an important risk factor for urinary tract infection in post-menopausal women.

When the case happens where you have a consistent occurrence of a UTI, some turn to a very last resort. This is called a bladder neck closure, which is where they remove and close up the urethra. However in a study done by (Wada Naoki, 2025) there was a case study done on a man with the disease. The results ended up being a failure. Not only was there no improvement on curing the disease, but it even made it worse. So this procedure can help, but also cause worse symptoms upon the original disease.

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UN-SDG TOPIC: Urinary Tract Infections In Women

Group 8 - Three Anonymous Students

College of Southern Nevada

Sociology 101

Professor Flora E. Rudacille

12/7/2025

Urinary tract infections (UTI) are among one of the most common bacterial infections worldwide, but their impact varies depending on gender, age, and social conditions. The causes and risks affect both men and women around the world. While women experience higher infection rates due to anatomical and hormonal factors, men often face delayed diagnosis or inappropriate treatment because UTIs are less common in the male population. However, social and environmental conditions such as access to clean water, menstrual hygiene, healthcare availability, and proper antibiotic use, UTI outcomes all genders. Research shows that UTIs become especially complex in populations facing poverty, aging, pregnancy, showing that these infections are not only biological conditions but also public health issues influenced by inequality and inconsistent medical practices. All together, the studies in men and women have shown the understanding that UTIs require when examining how gender, environment, and healthcare systems interact to impact infection risk and treatment success.

Environmental and social inequalities significantly shape UTI risk for both men and women, though women are excessively affected. In low income communities, limited to clean water and sanitation increases infection risk by making basic hygiene difficult for everyone, but most specifically to women who do require additional hygiene support during menstruation. Makusha (2020) stated that, “inadequate access to clean water and sanitation in informal settlements increases the vulnerability of female residents to UTIs,” showing how environmental conditions affect women's health. Menstruators facing financial challenges also experience increased infection risk because they often rely on unsafe materials during their periods, which can lead to bad bacteria and heighten the higher chance of an infection (Glazer, Jennings, Schlaeger, Watkins, Rieseler, Ray, Lee, & Glazer, 2024). Men as well are affected by the environmental and systemic issues mostly when limited to healthcare it leads to delayed diagnosis or misuse of antibiotics. In Malawi, for example, nearly half of UTI cases in both men and women have shown resistance to common antibiotics due to poor hygiene and self medication (Pensulo, 2025). These studies have shown that while UTIs impact all genders, social inequality and as well environmental barriers, that makes UTI a public health concern rather than just a biological condition.

Research further demonstrates how age, health care quality, and treatment practices shape UTI outcomes across populations. Pregnant individuals face uniquely high risk due to physiological changes that change urinary flow and weaken immune responses. In South Africa, research found that pregnant women admitted with UTIs often experienced complications such as preterm birth, anemia, and kidney impairment, especially in hospitals where follow-up care and diagnostic accuracy were inconsistent (de Paul, 2016). These findings underscore how UTI severity escalates when vulnerable groups lack timely, high-quality medical attention. Similarly, older adults, particularly older women, face heightened risk due to age-related changes, weaker immune systems, and other comorbidities. While some theories have linked older adults infection risk to gut microbiome disruptions, (Winter, Obara, & McMahon, 2020) found no significant connections between microbiome variations and UTI history in aged-care residents. Instead, their research suggested that factors such as antibiotic exposure, chronic illness, and environmental conditions may play far greater roles in infection susceptibility.

Research shows that preventing and treating UTIs involves more than just taking antibiotics, especially for women who get them more frequently. “It is estimated at least one third of women will develop a urinary tract infection (UTI) in their lifetime” (Davis & Rantell, 2017). Many women get repeated infections because bacteria from the gut can reach the urinary tract. This risk is higher when hygiene is difficult, such as not having access to clean water, using unsafe menstrual products, or not having proper washing facilities. These findings show that stopping UTIs requires not only medicine but also education, better hygiene, and access to clean water. It highlights that UTIs are affected by both biology and social conditions. Addressing these factors can help reduce the number of infections and prevent serious health complications. Community programs that teach proper hygiene and provide resources for safe menstrual care have been shown to lower UTI rates in vulnerable populations.

Beyond environmental and social factors, research has shown how the treatment quality and antibiotic management played a major role in shaping UTI outcomes, especially for men and older women. While UTIs are less common in men, inappropriate or

inconsistent antibiotic prescribing is still a main issue. (Biguenet et al, 2025) found that male UTI patients often received antibiotics that did not align with clinical recommendations, increasing the risk of resistance and treatment failure. These prescribing patterns highlight how gaps in healthcare knowledge and diagnostic accuracy can worsen infection outcomes for men and make their health worsen. Older women, however, face recurring UTIs that are influenced not only by biological changes but as well by medical system challenges. According to (Gani dağlı, Öztürk, and Öztürk 2023), recurrent UTIs in older women are often tied to poor treatment plans, delayed medical evaluation, and the underrecognition of early symptoms, making the condition more complex than just a simple infection. Together, these findings reveal that UTI outcomes depend heavily on the quality of healthcare access and antibiotics, emphasizing the need for more consistent, evidence based treatment across genders and age groups.

In conclusion, UTIs are a common health problem that can affect anyone, but age, gender, and living conditions make a big difference in how serious they are. Women tend to get UTIs more often, but men, older adults, and pregnant women can also face serious 5 complications, especially if they don't have easy access to healthcare. Things like not having clean water, safe menstrual products, or proper bathrooms can make infections more likely. Treating UTIs isn't just about taking medicine, it also means learning about hygiene, having the right resources and supporting people in the community. By paying attention to both the medical and social side of the problem, we can help prevent infections and keep people healthier.

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REFLECTION

My name is Flora Rudacille, and I teach Sociology in the Human Behavior Department at the College of Southern Nevada in Las Vegas, NV. During the fall semester, 2025, I had the opportunity to collaborate with Dr. Catherine Glass, Biology professor in the Health Sciences Department at Langara College on a project in the UNSDG Fellowship program. Dr. Glass and I developed three scaffolded renewable assignments connected to UNSDG Goal #5, Gender Equality, that explore the associations between disease and gender and other socioeconomic and demographic factors. The final product of the assignments is a research paper exploring a particular disease and its associations with these factors.

Students worked in groups in two of my Sociology 101 classes. Each group consisted of three or four students who worked on all three assignments together, a project proposal and preliminary references, an annotated bibliography and a final research paper/case study which would be included in an open resource undergraduate minitext.

I worked closely with our Henderson Campus librarian, and UNSDG Mentor/Program Lead, Ms. Gracie McDonough, throughout the semester to assist students in locating high quality resources, developing their annotated bibliographies and final research papers in APA format. I was very fortunate to have Ms. McDonough on my campus and made for a much richer educational experience for my students. In addition, our Center for Academic Success student support services department also provided support to our student groups over the course of the semester.

The students were very excited about the project and the opportunity to contribute to an open educational resource that would be available to other students such as themselves, and the students remained engaged throughout the semester. They learned how to locate quality references and resources; how to do an annotated bibliography and how to write a research paper in APA format; along with how to apply theory, make connections between a real-world disease or condition and variety of socioeconomic factors, and to consider strategies needed to address the issues discovered. We ended with thirteen papers from my Sociology 101 classes for the mini text, all of which were done very well by first year undergraduate students.

In terms of the work that it took on my end, it took a bit more than usual due to implementation of the new UNSDG projects and deeper collaboration with our library and student support services staff, but this work was highly rewarding in a variety of ways for me and for my students. In fact, it was so rewarding that I am doing the project again in the spring semester.

I expect the “Sociodemics” mini text to be available in the very near future and I hope you will take a look at the work done by these students!

Submitted by Prof. Flora Rudacille, 1/15/2026

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